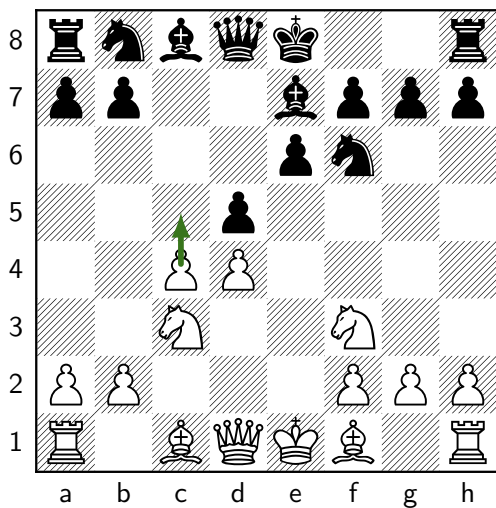


Melanie-BASE — Caro-Kann, Panov

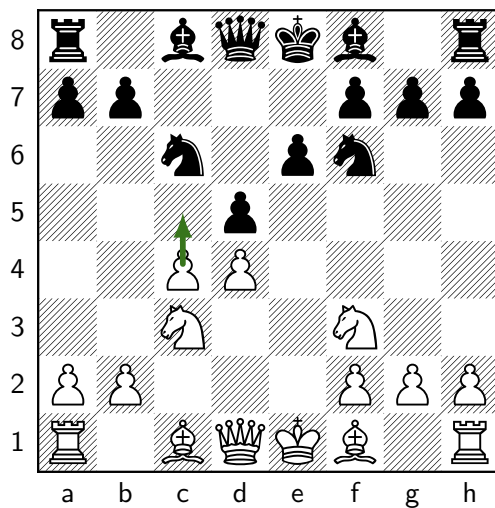
Melanie (Deep_Vision)

Abstract

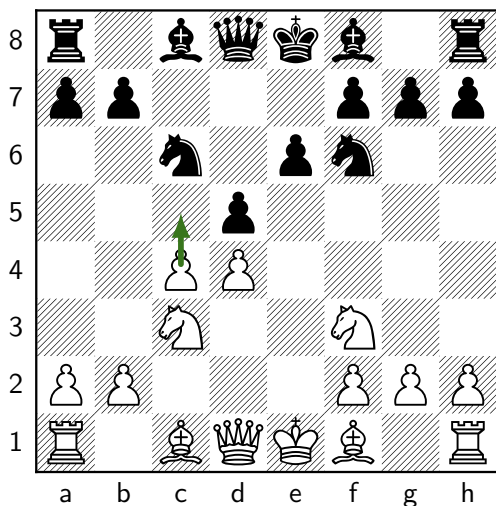
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



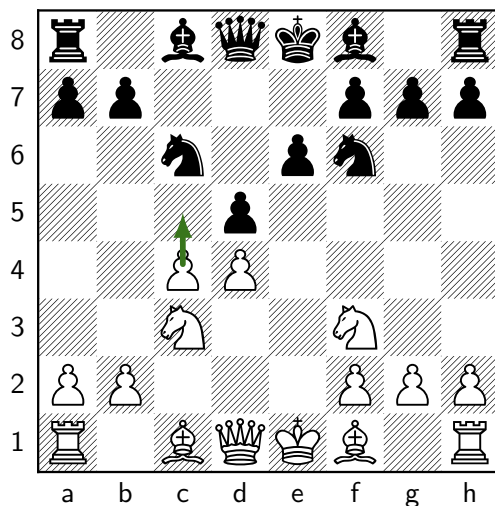
(a)



(b)

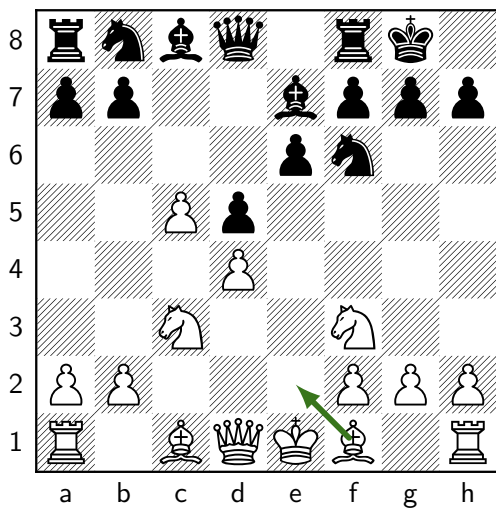


(c)

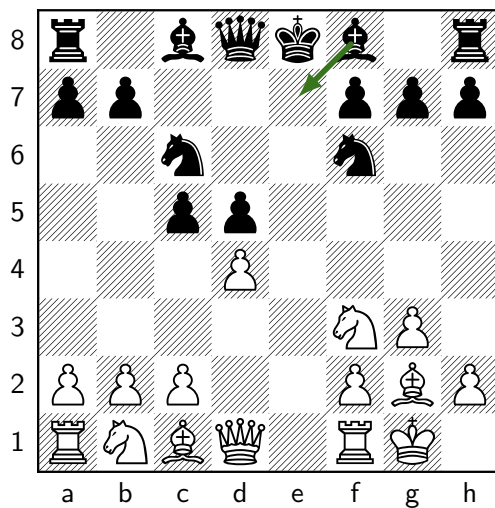


(d)

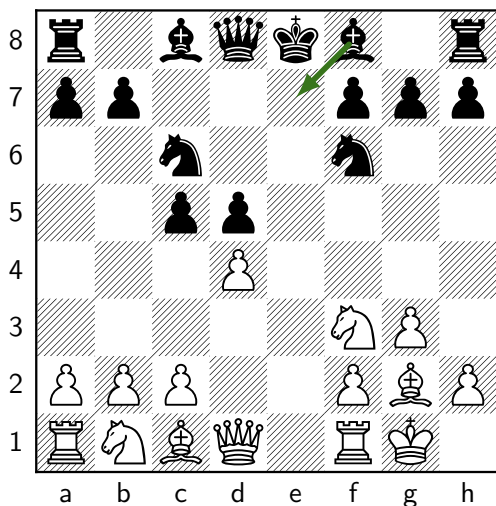
Figure 1: **Move 7, White: c5.** (a) Our position; Melanie plays c5 (c4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3405` (ply 12) [structural distance $d=29$, QAP obj=109]: the master played c5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Pb2$ as their $Bf1$, our $be7$ as their $Pb2$; transferred move c4c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3406` (ply 12) [structural distance $d=29$, QAP obj=109]: the master played c5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Pb2$ as their $Bf1$, our $be7$ as their $Pb2$; transferred move c4c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#180` (ply 12) [structural distance $d=29$, QAP obj=109]: the master played c5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Pb2$ as their $Bf1$, our $be7$ as their $Pb2$; transferred move c4c5 (**endorses this move**). Synthesis: 21 retrieved analogs transferred to c5 (summed QAP objective 1850, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



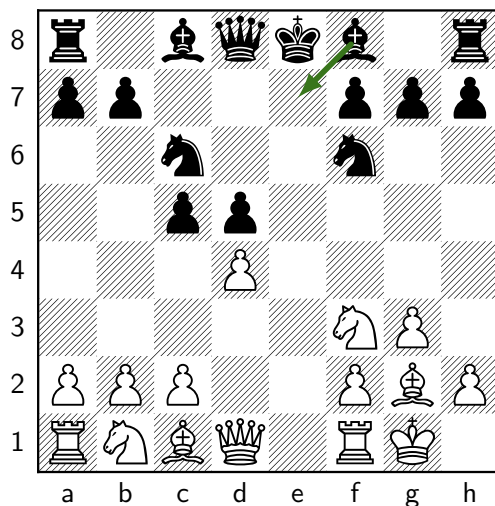
(a)



(b)

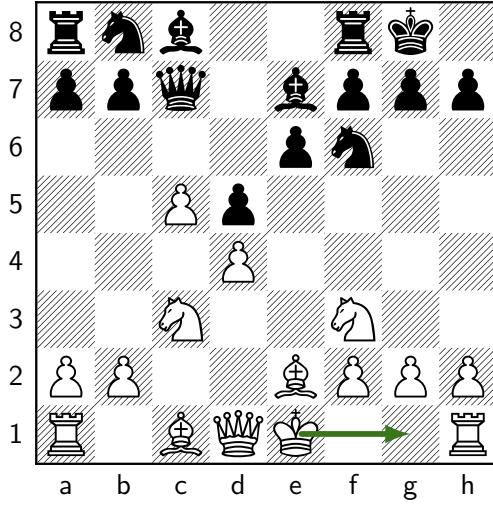


(c)

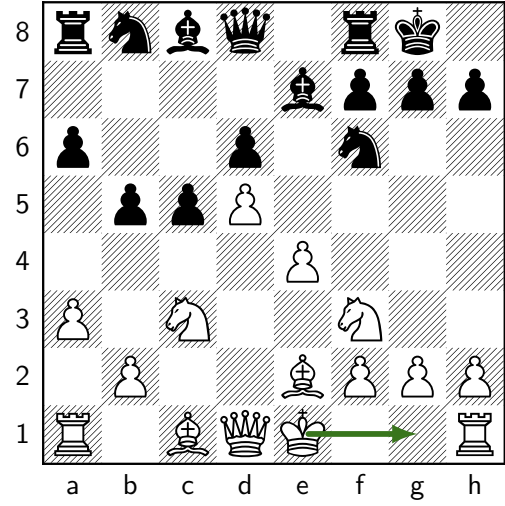


(d)

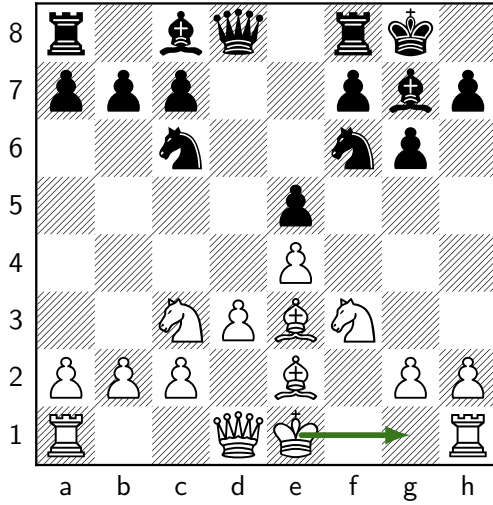
Figure 2: **Move 8, White: Be2.** (a) Our position; Melanie plays Be2 (f1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#876 (ply 13) [structural distance $d=36$, QAP obj=30]: the master played Be7; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our Pd4 as their Bc8, our Bc1 as their Ph7; transferred move f1e2 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#877 (ply 13) [structural distance $d=36$, QAP obj=30]: the master played Be7; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our Pd4 as their Bc8, our Bc1 as their Ph7; transferred move f1e2 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#2505 (ply 13) [structural distance $d=36$, QAP obj=30]: the master played Be7; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our Pd4 as their Bc8, our Bc1 as their Ph7; transferred move f1e2 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Be2 (summed QAP objective 282, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



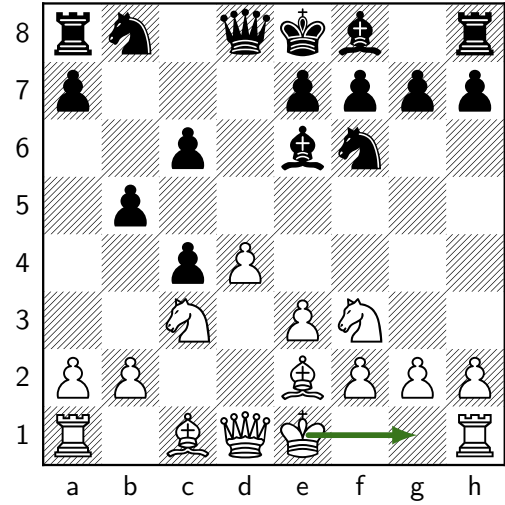
(a)



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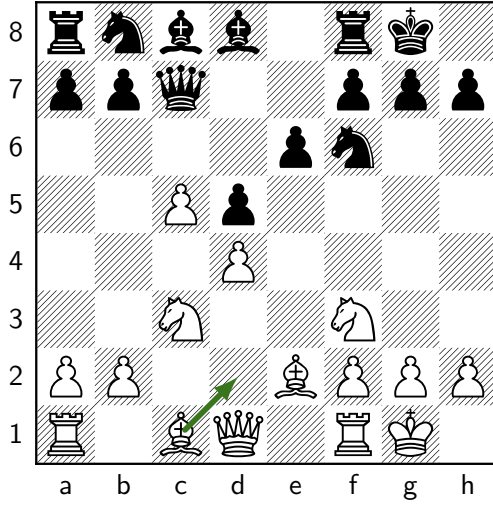


(c)

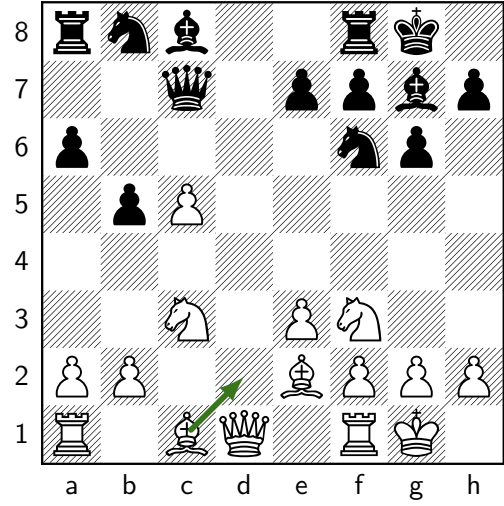


(d)

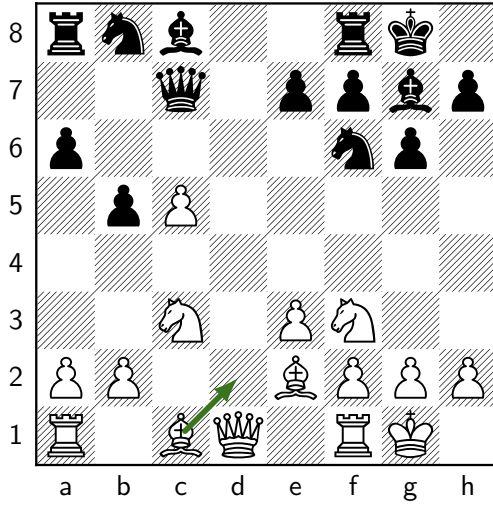
Figure 3: **Move 9, White: 0-0 (e1g1).** (a) Our position; Melanie plays 0-0 (e1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3547` (ply 20) [structural distance $d=51$, QAP obj=72]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Nf3* as their *Rh1*, our *Bc1* as their *Pg2*; transferred move e1g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3914` (ply 16) [structural distance $d=55$, QAP obj=62]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Be2* as their *Ra1*, our *Nf3* as their *Qd1*; transferred move e1g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#300` (ply 12) [structural distance $d=56$, QAP obj=59]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Ra1*, our *nb8* as their *Pa2*; transferred move e1g1 (**endorses this move**). Synthesis: 24 retrieved analogs transferred to 0-0 (summed QAP objective 1150, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



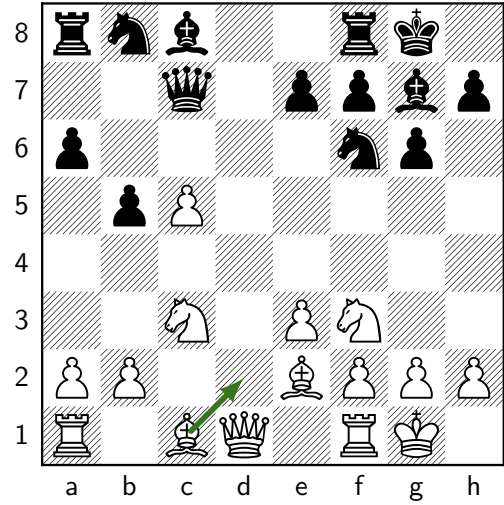
(a)



(b)

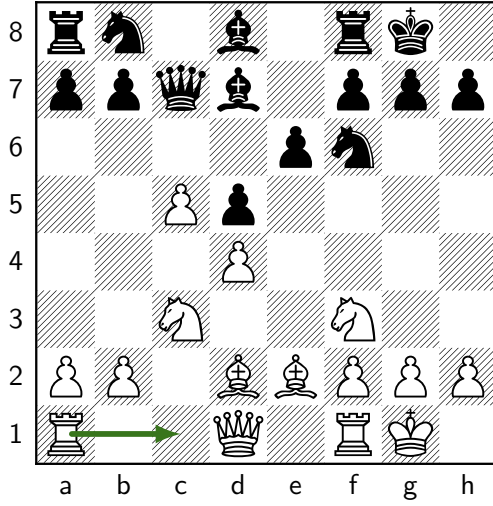


(c)

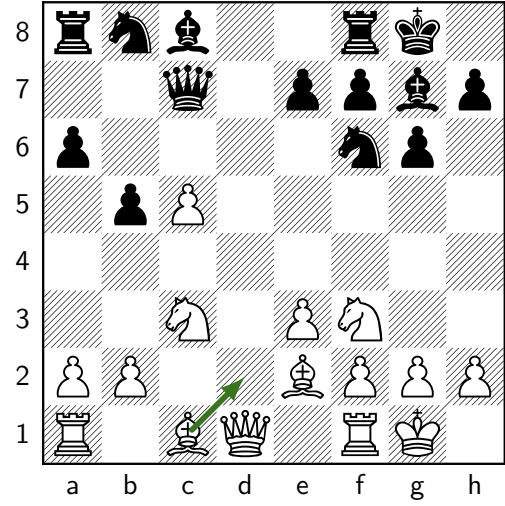


(d)

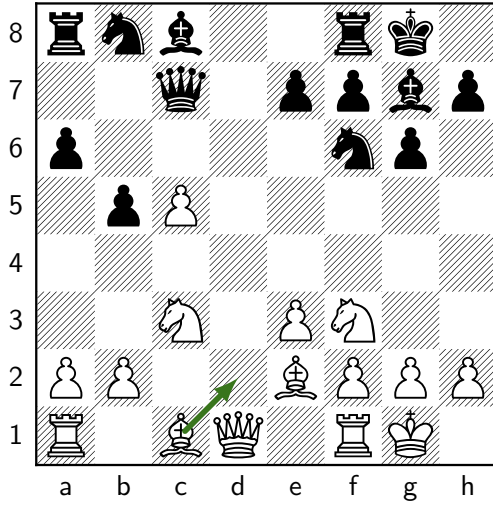
Figure 4: **Move 10, White: Bd2.** (a) Our position; Melanie plays Bd2 (c1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3819` (ply 20) [structural distance $d=54$, QAP obj=94]: the master played Bd2; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *Be2* as their *Qd1*, our *Nc3* as their *Be2*; transferred move c1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1878` (ply 20) [structural distance $d=54$, QAP obj=94]: the master played Bd2; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *Be2* as their *Qd1*, our *Nc3* as their *Be2*; transferred move c1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1879` (ply 20) [structural distance $d=54$, QAP obj=94]: the master played Bd2; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *Be2* as their *Qd1*, our *Nc3* as their *Be2*; transferred move c1d2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bd2 (summed QAP objective 282, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



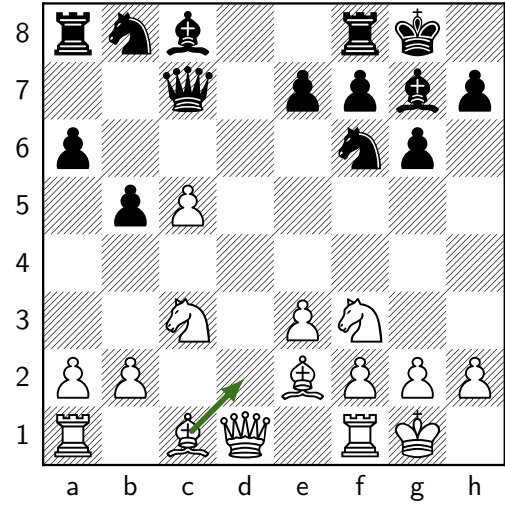
(a)



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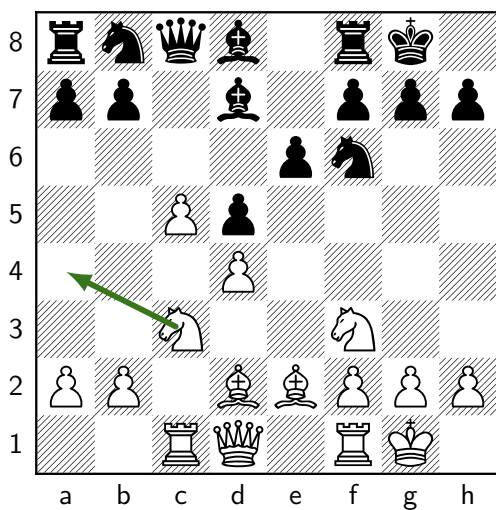


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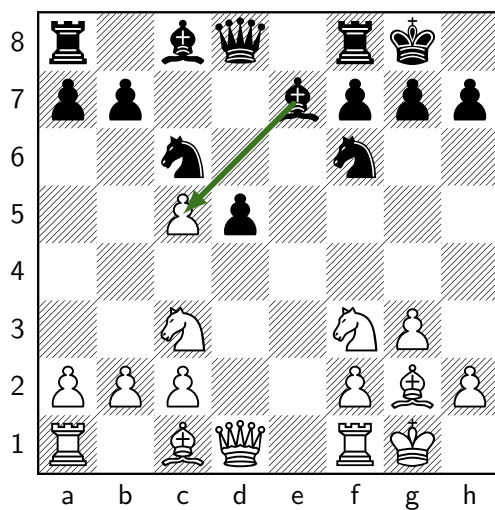


(d)

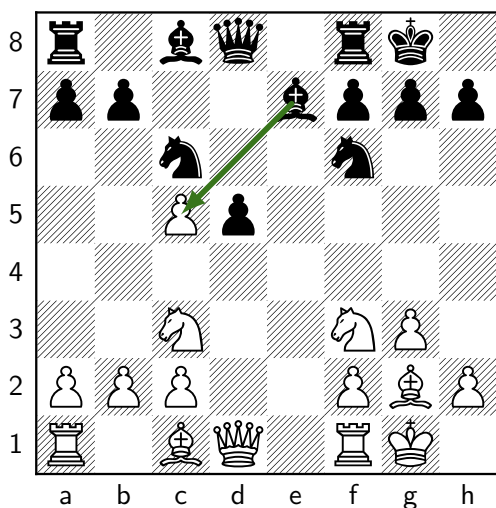
Figure 5: **Move 11, White: Rc1.** (a) Our position; Melanie plays Rc1 (a1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3819` (ply 20) [structural distance $d=59$, QAP obj=95]: the master played Bd2; under μ our *Ra1* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Ra1*, our *Be2* as their *Qd1*; transferred move a1c1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1878` (ply 20) [structural distance $d=59$, QAP obj=95]: the master played Bd2; under μ our *Ra1* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Ra1*, our *Be2* as their *Qd1*; transferred move a1c1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1879` (ply 20) [structural distance $d=59$, QAP obj=95]: the master played Bd2; under μ our *Ra1* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Ra1*, our *Be2* as their *Qd1*; transferred move a1c1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rc1 (summed QAP objective 284, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



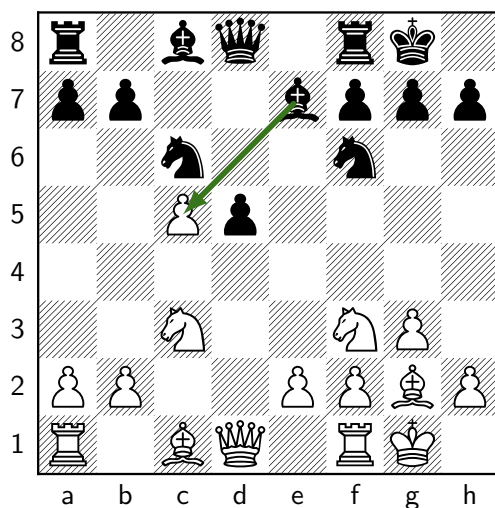
(a)



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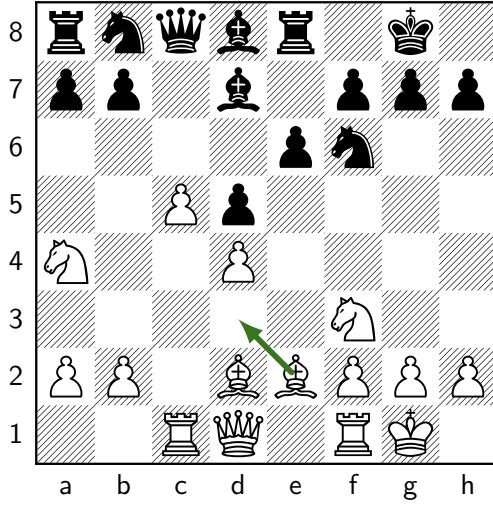


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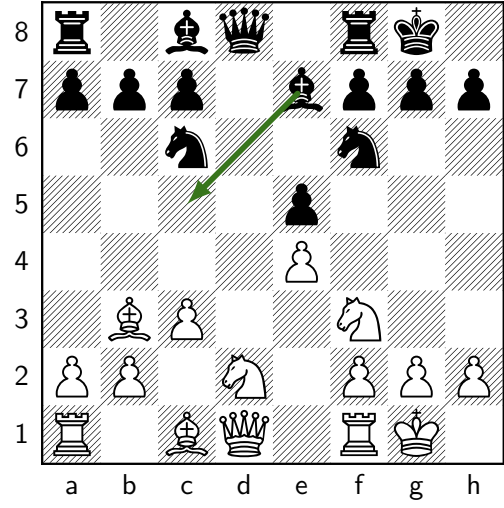


(d)

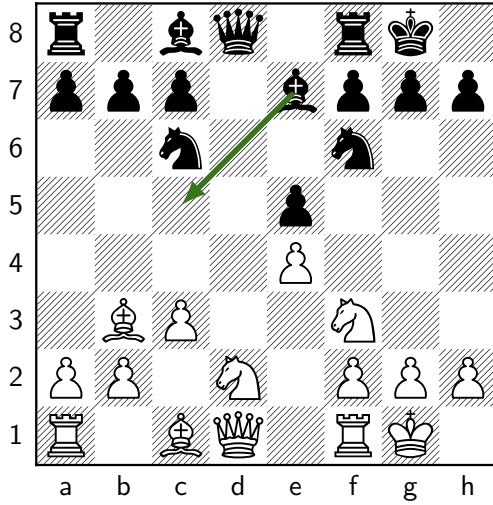
Figure 6: **Move 12, White: Na4.** (a) Our position; Melanie plays Na4 (c3a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#877` (ply 17) [structural distance $d=74$, QAP obj=69]: the master played Bxc5; under μ our Nc3 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our Be2 as their Ra8, our Pd4 as their Qd8; transferred move c3a4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2635` (ply 17) [structural distance $d=74$, QAP obj=69]: the master played Bxc5; under μ our Nc3 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our Be2 as their Ra8, our Pd4 as their Qd8; transferred move c3a4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#101` (ply 17) [structural distance $d=75$, QAP obj=67]: the master played Bxc5; under μ our Nc3 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our nf6 as their Pb7, our Pd4 as their Nc6; transferred move c3a4 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Na4 (summed QAP objective 1070, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



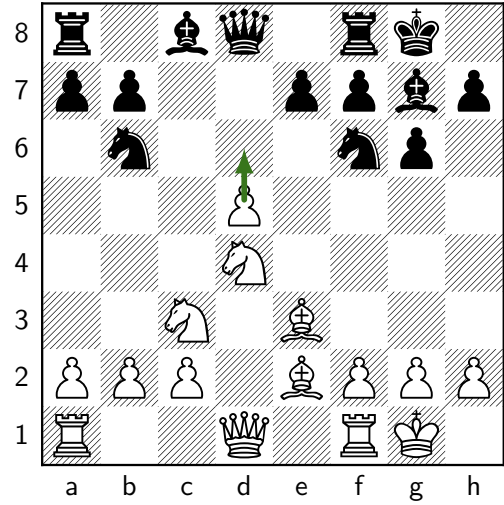
(a)



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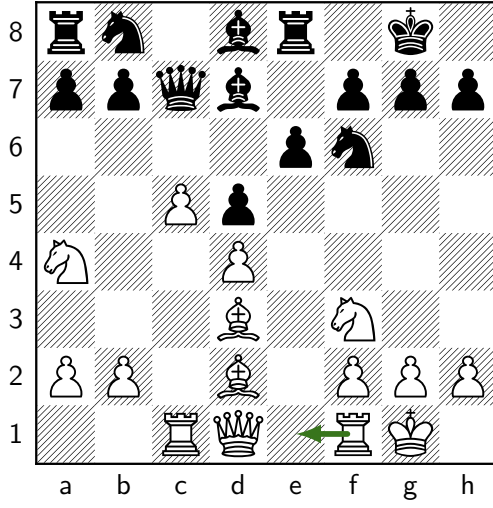


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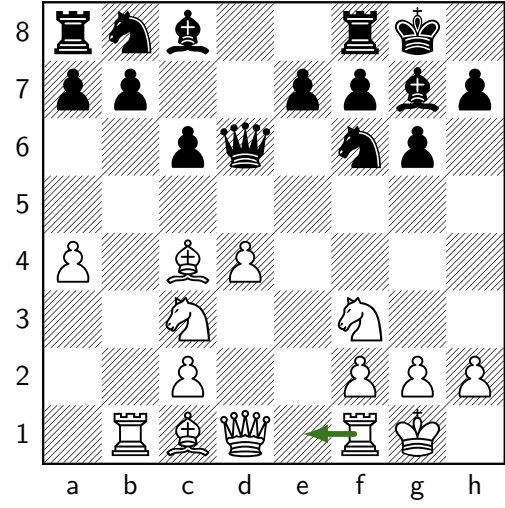


(d)

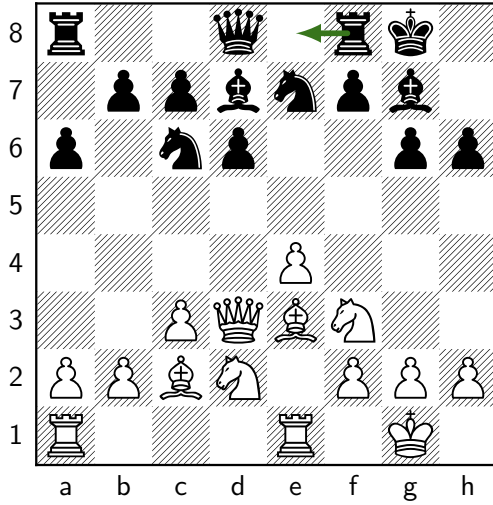
Figure 7: **Move 13, White: Bd3.** (a) Our position; Melanie plays Bd3 (e2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2243` (ply 17) [structural distance $d=58$, QAP obj=43]: the master played Bc5; under μ our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *pd5* as their *Ra8*, our *Nf3* as their *Rf8*; transferred move e2d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2244` (ply 17) [structural distance $d=58$, QAP obj=43]: the master played Bc5; under μ our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *pd5* as their *Ra8*, our *Nf3* as their *Rf8*; transferred move e2d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2247` (ply 20) [structural distance $d=59$, QAP obj=52]: the master played d6; under μ our *Pc5* plays the role of that game’s *Pd5*; cross-type roles: our *pg7* as their *Qd1*, our *nf6* as their *Pb2*; transferred move c5c6. Synthesis: 2 retrieved analogs transferred to Bd3 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



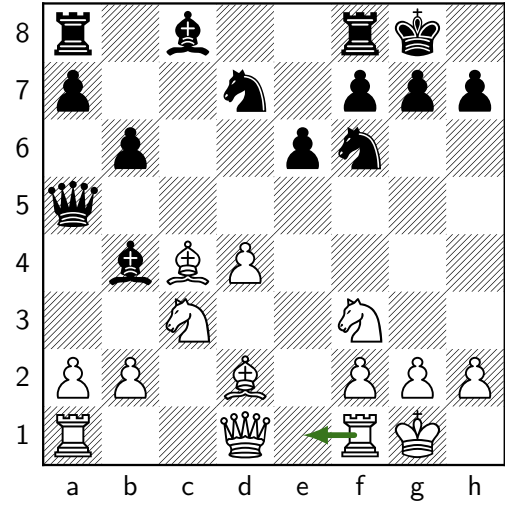
(a)



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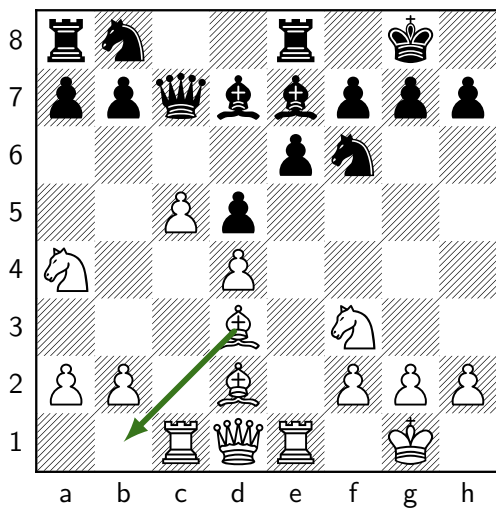


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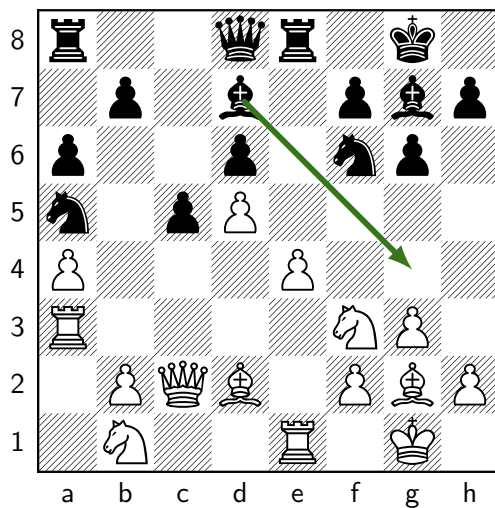


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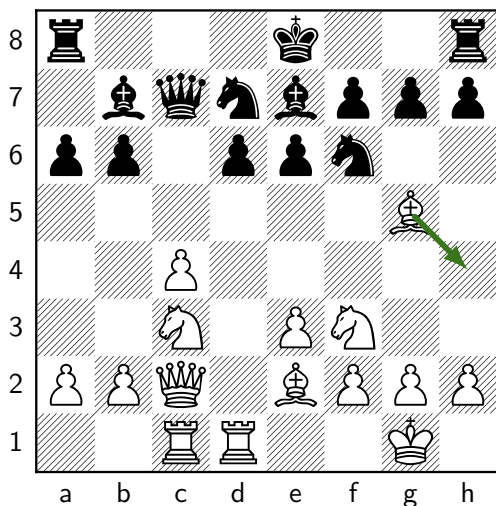
Figure 8: **Move 14, White: Re1.** (a) Our position; Melanie plays Re1 (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1545` (ply 20) [structural distance $d=77$, QAP obj=60]: the master played Re1; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Bd3* as their *Rb1*, our *Nf3* as their *Qd1*; transferred move f1e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1058` (ply 25) [structural distance $d=68$, QAP obj=49]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Ph2* as their *Bd7*, our *Pd4* as their *Ne7*; transferred move f1e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#970` (ply 22) [structural distance $d=65$, QAP obj=37]: the master played Re1; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Bd3* as their *Qd1*, our *pe6* as their *Bc4*; transferred move f1e1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Re1 (summed QAP objective 176, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



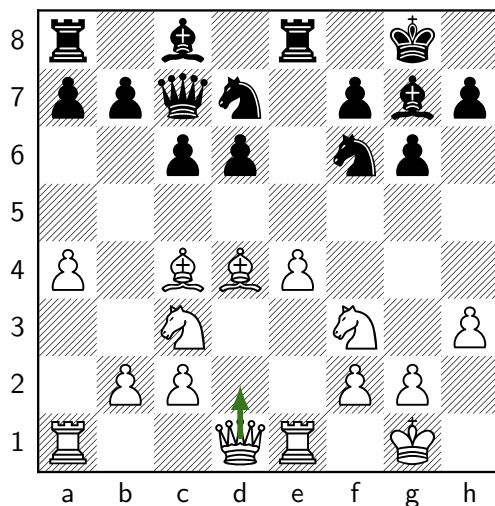
(a)



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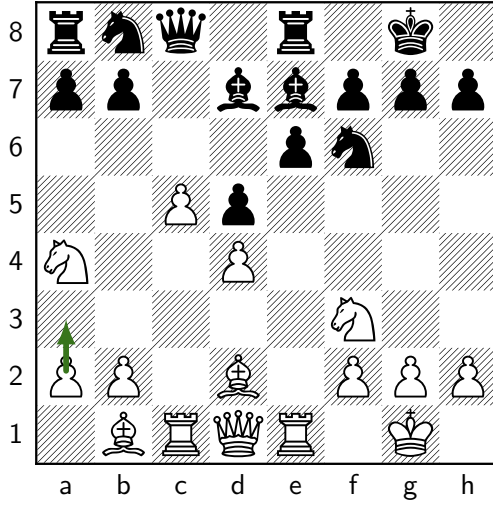


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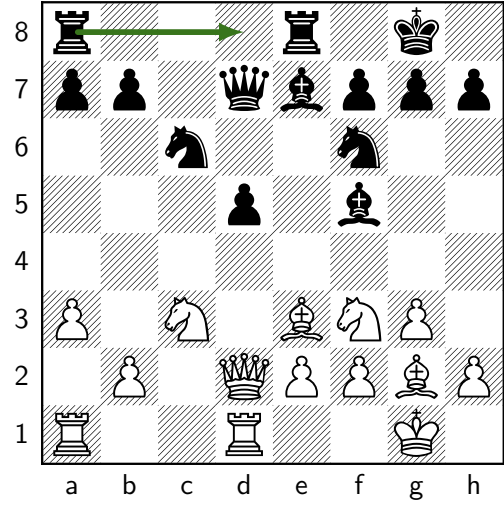


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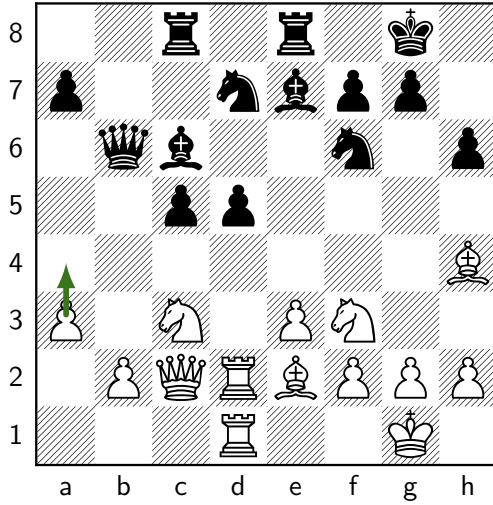
Figure 9: **Move 15, White: Bb1.** (a) Our position; Melanie plays Bb1 (d3b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0001#2630` (ply 27) [structural distance $d=82$, QAP obj=63]: the master played Bg4; under μ our Bd3 plays the role of that game’s Bd2; cross-type roles: our Qd1 as their Ra8, our Bd2 as their Qd8; transferred move d3b1 (**endorses this move**). (c) Retrieved from `cctl_top_p0000#3405` (ply 24) [structural distance $d=96$, QAP obj=40]: the master played Bh4; under μ our Bd3 plays the role of that game’s Bg5; cross-type roles: our Ph2 as their Rc1, our Rc1 as their Pa2; transferred move d3b1 (**endorses this move**). (d) Retrieved from `cctl_top_p0000#2711` (ply 26) [structural distance $d=89$, QAP obj=40]: the master played Qd2; under μ our Bd3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Ra1, our Nf3 as their Re1; transferred move d3b1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bb1 (summed QAP objective 143, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



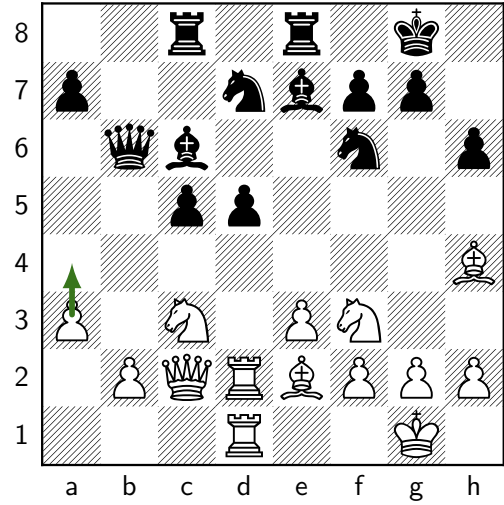
(a)



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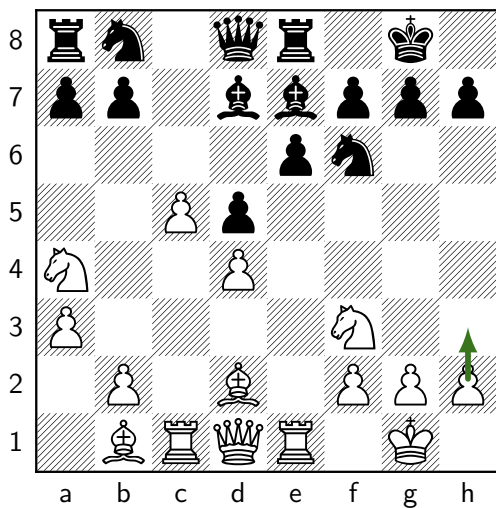


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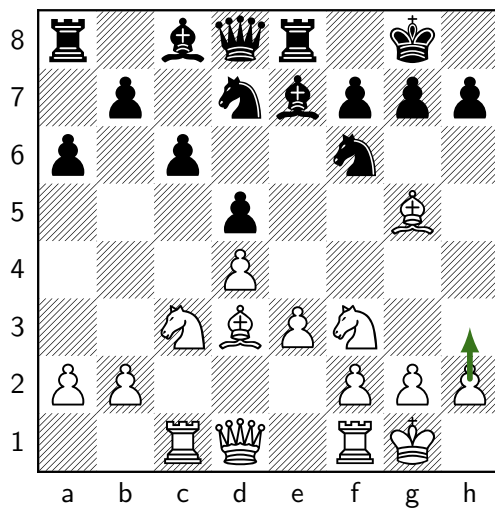


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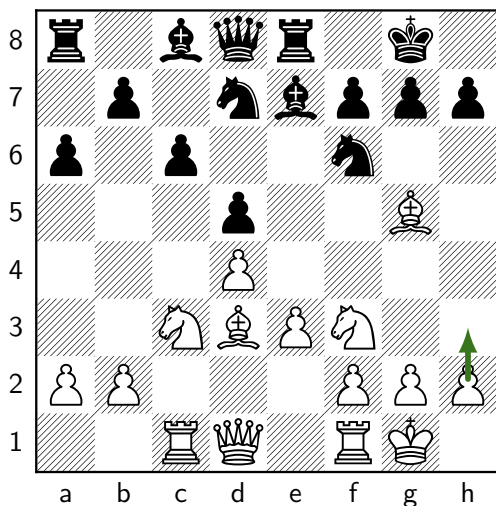
Figure 10: **Move 16, White: a3.** (a) Our position; Melanie plays a3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2637` (ply 27) [structural distance $d=73$, QAP obj=74]: the master played Rad8; under μ our Pa2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Na4 as their Pa7, our Rc1 as their Qd7; transferred move a2a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#546` (ply 32) [structural distance $d=73$, QAP obj=69]: the master played a4; under μ our Pa2 plays the role of that game’s Pa3; cross-type roles: our Rc1 as their Pb2, our pb7 as their Qc2; transferred move a2a3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#692` (ply 32) [structural distance $d=73$, QAP obj=69]: the master played a4; under μ our Pa2 plays the role of that game’s Pa3; cross-type roles: our Rc1 as their Pb2, our pb7 as their Qc2; transferred move a2a3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to a3 (summed QAP objective 261, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



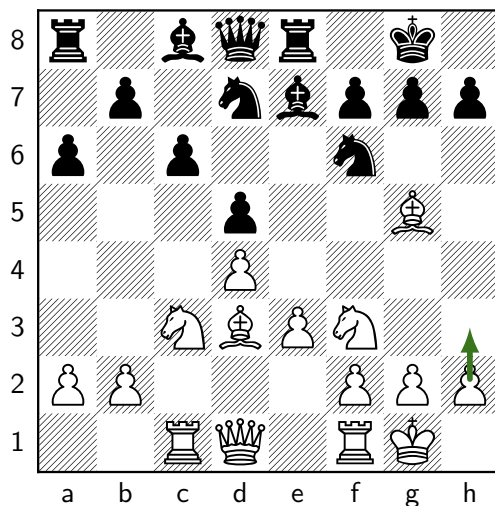
(a)



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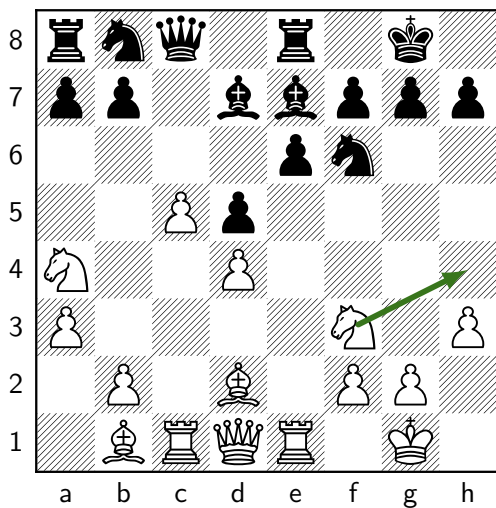


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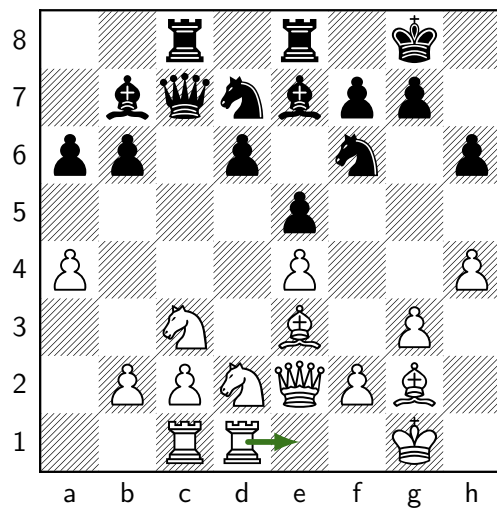


(d)

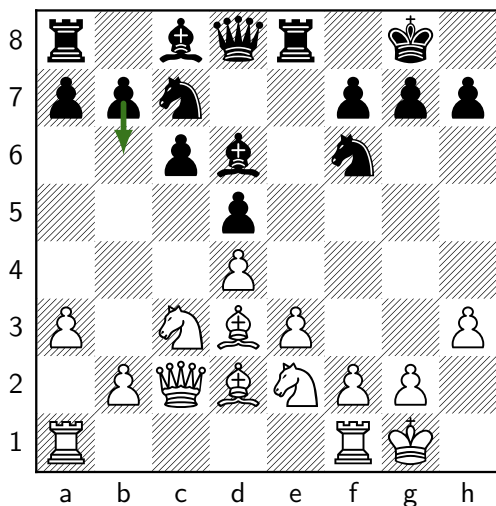
Figure 11: **Move 17, White: h3.** (a) Our position; Melanie plays h3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1171` (ply 20) [structural distance $d=61$, QAP obj=13]: the master played h3; under μ our $Ph2$ plays the role of that game’s $Ph2$; cross-type roles: our $Bb1$ as their $Rc1$, our $pg7$ as their $Qd1$; transferred move h2h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1517` (ply 20) [structural distance $d=61$, QAP obj=13]: the master played h3; under μ our $Ph2$ plays the role of that game’s $Ph2$; cross-type roles: our $Bb1$ as their $Rc1$, our $pg7$ as their $Qd1$; transferred move h2h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3613` (ply 20) [structural distance $d=61$, QAP obj=13]: the master played h3; under μ our $Ph2$ plays the role of that game’s $Ph2$; cross-type roles: our $Bb1$ as their $Rc1$, our $pg7$ as their $Qd1$; transferred move h2h3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to h3 (summed QAP objective 76, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



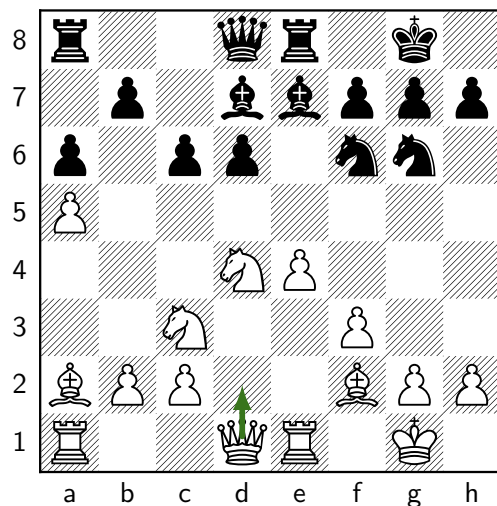
(a)



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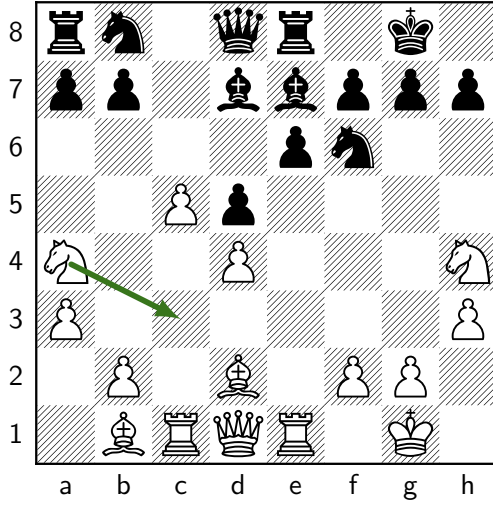


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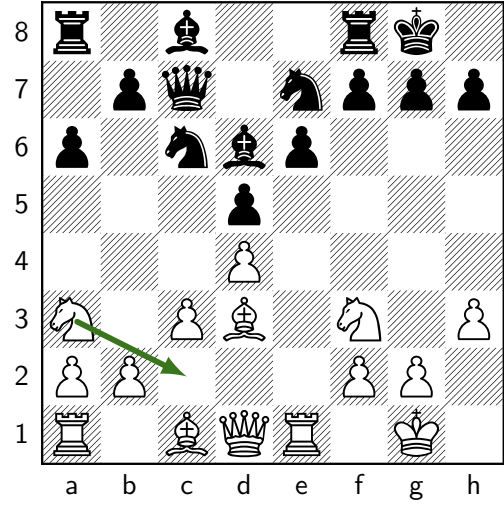


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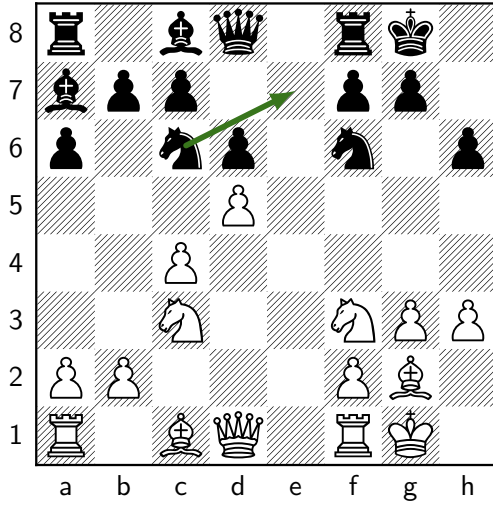
Figure 12: **Move 18, White: Nh4**. (a) Our position; Melanie plays Nh4 (f3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3761` (ply 44) [structural distance $d=70$, QAP obj=66]: the master played R $e1$; under μ our Nf3 plays the role of that game’s R $d1$ (a **cross-type** correspondence); cross-type roles: our Q $d1$ as their R $c1$, our P $d4$ as their N $d2$; transferred move f3h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1916` (ply 23) [structural distance $d=60$, QAP obj=40]: the master played b6; under μ our Nf3 plays the role of that game’s P $b2$ (a **cross-type** correspondence); cross-type roles: our p $d5$ as their R $a8$, our P $f2$ as their B $c8$; transferred move f3h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#801` (ply 28) [structural distance $d=73$, QAP obj=40]: the master played Q $d2$; under μ our Nf3 plays the role of that game’s Q $d1$ (a **cross-type** correspondence); cross-type roles: our Q $d1$ as their R $a1$, our P $h3$ as their R $e1$; transferred move f3h4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nh4 (summed QAP objective 183, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



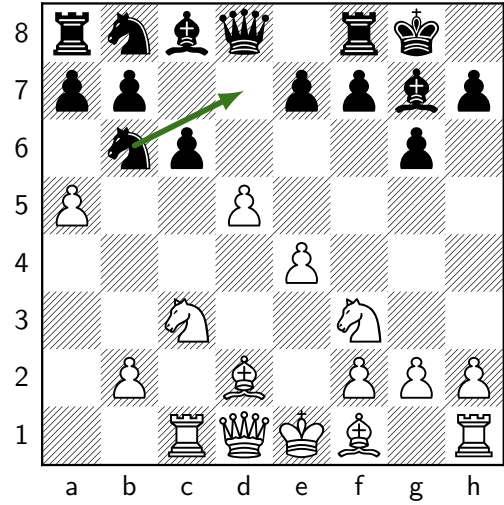
(a)



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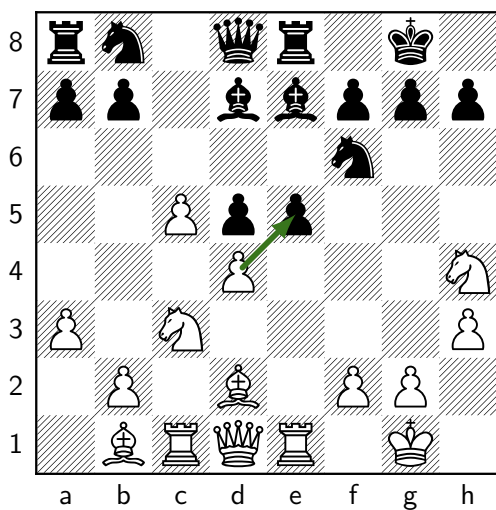


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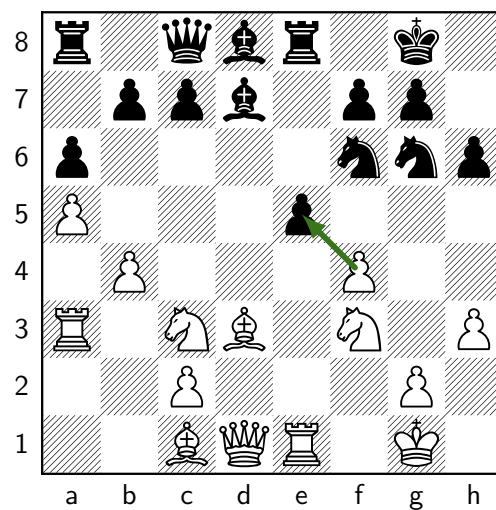


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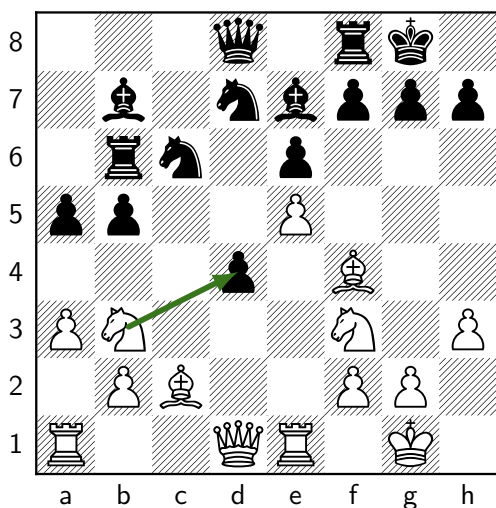
Figure 13: **Move 19, White: Nc3.** (a) Our position; Melanie plays Nc3 (a4c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3498` (ply 20) [structural distance $d=76$, QAP obj=61]: the master played Nc2; under μ our Na4 plays the role of that game’s Na3; cross-type roles: our Nh4 as their Ra1, our Bd2 as their Pf2; transferred move a4c3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1667` (ply 21) [structural distance $d=75$, QAP obj=35]: the master played Ne7; under μ our Na4 plays the role of that game’s Nc3; cross-type roles: our Qd1 as their Ra8, our Ph3 as their Bc8; transferred move a4c3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#587` (ply 21) [structural distance $d=69$, QAP obj=29]: the master played N6d7; under μ our Na4 plays the role of that game’s Nb3; cross-type roles: our be7 as their Ra8, our ph7 as their Nb8; transferred move a4c3 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Nc3 (summed QAP objective 225, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



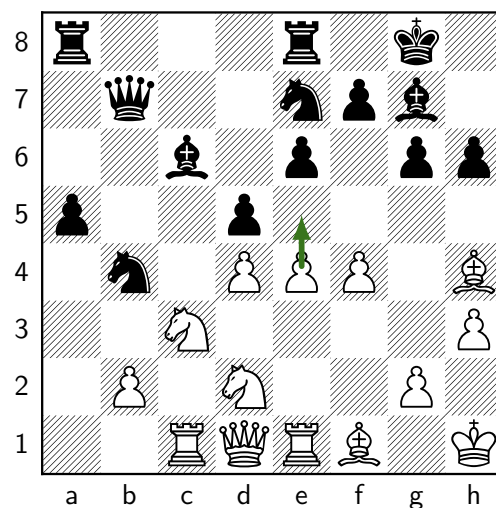
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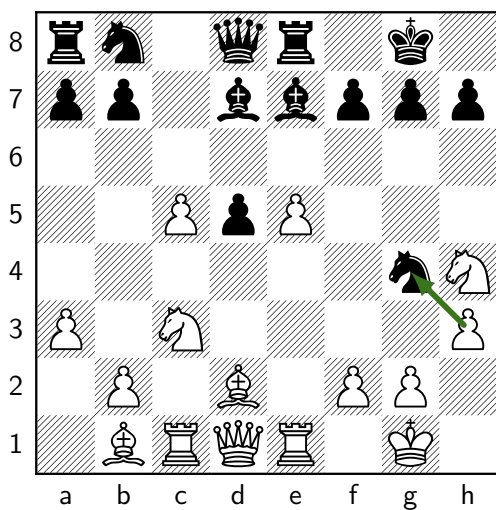


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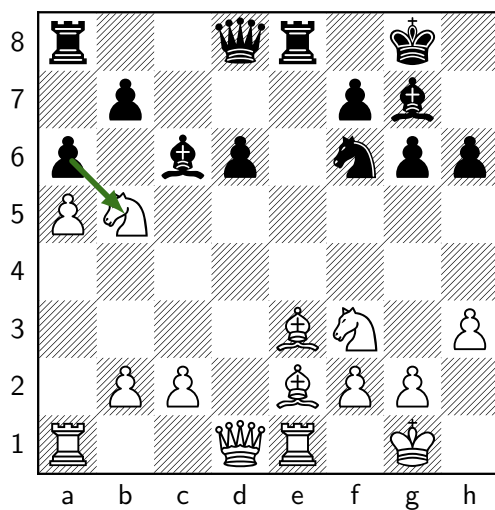


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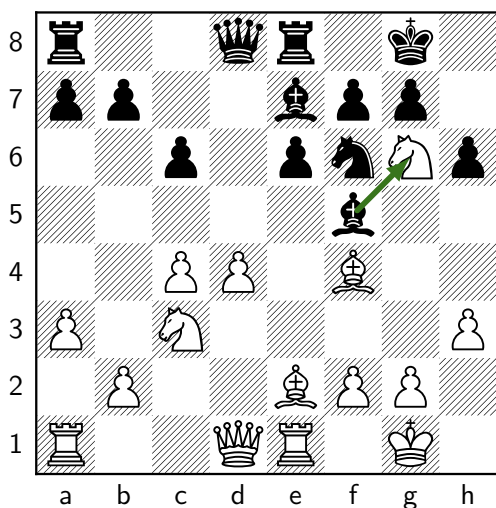
Figure 14: **Move 20, White: dxe5.** (a) Our position; Melanie plays dxe5 (d4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1335` (ply 36) [structural distance $d=97$, QAP obj=155]: the master played fxe5; under μ our $Pd4$ plays the role of that game’s $Pf4$; cross-type roles: our $Bb1$ as their $Nc3$, our $Pd4$ as their $Bd3$; transferred move d4e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#721` (ply 32) [structural distance $d=98$, QAP obj=77]: the master played Nbx d4; under μ our $Pd4$ plays the role of that game’s $Nb3$ (a **cross-type** correspondence); cross-type roles: our $Qd1$ as their $Ra1$, our $Re1$ as their $Qd1$; transferred move d4e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#26` (ply 44) [structural distance $d=101$, QAP obj=74]: the master played e5; under μ our $Pd4$ plays the role of that game’s $Pe4$; cross-type roles: our $Bd2$ as their $Rc1$, our $bd7$ as their $Qd1$; transferred move d4e5 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to dxe5 (summed QAP objective 828, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



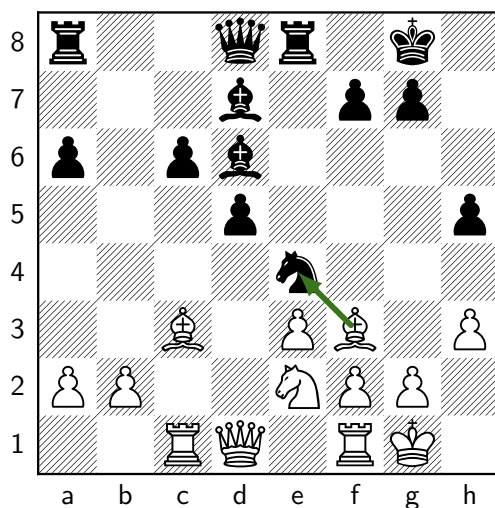
(a)



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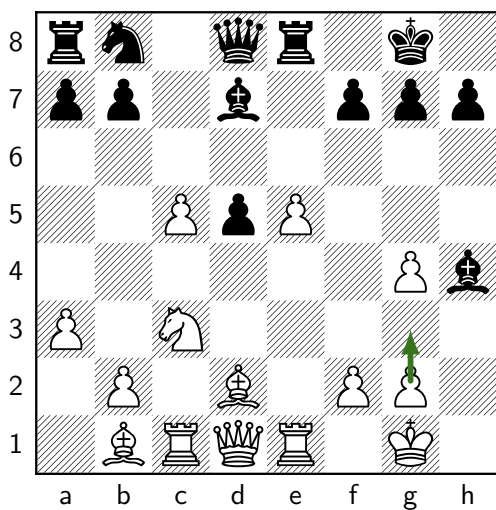


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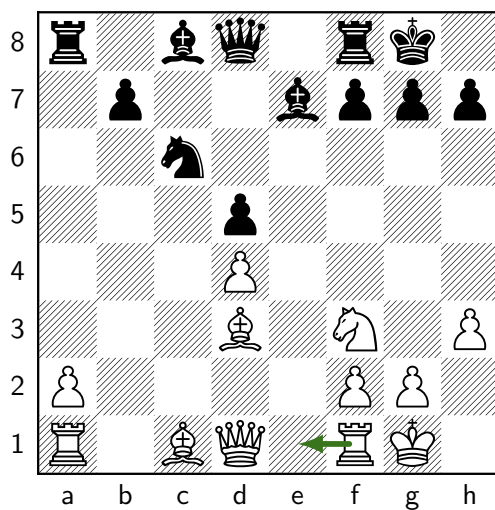


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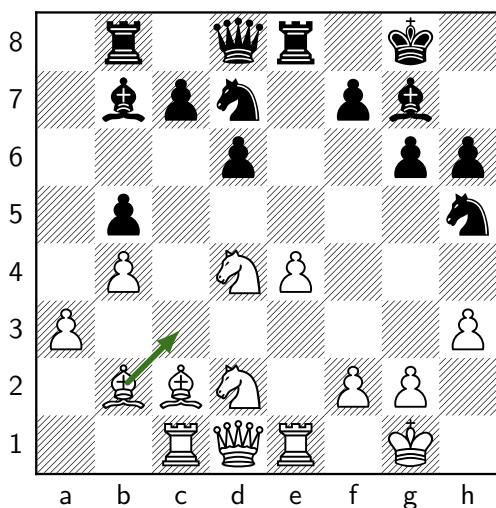
Figure 15: **Move 21, White: h4g4**. (a) Our position; Melanie plays h4g4 (h3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3343 (ply 35) [structural distance $d=106$, QAP obj=58]: the master played axb5; under μ our Ph3 plays the role of that game’s Pa3; cross-type roles: our Nc3 as their Ra8, our Bd2 as their Qd8; transferred move h3g4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3115 (ply 31) [structural distance $d=106$, QAP obj=43]: the master played Bxg6; under μ our Ph3 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our pd5 as their Ra8, our Bd2 as their Qd8; transferred move h3g4 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2024 (ply 32) [structural distance $d=83$, QAP obj=42]: the master played Bxe4; under μ our Ph3 plays the role of that game’s Bf3 (a **cross-type** correspondence); cross-type roles: our Bb1 as their Rc1, our Bd2 as their Pb2; transferred move h3g4 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to h4g4 (summed QAP objective 538, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



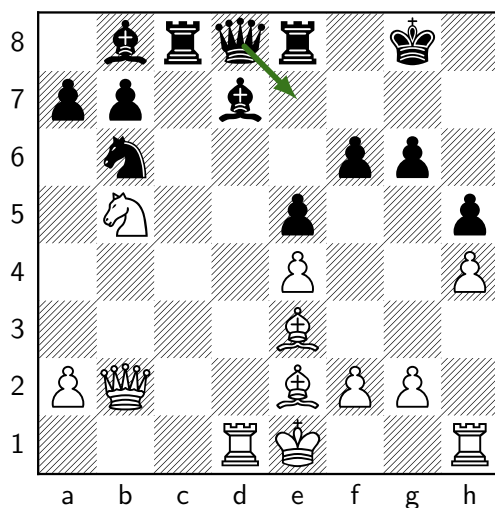
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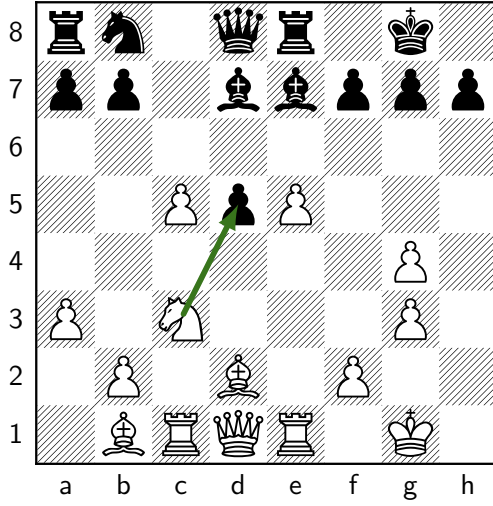


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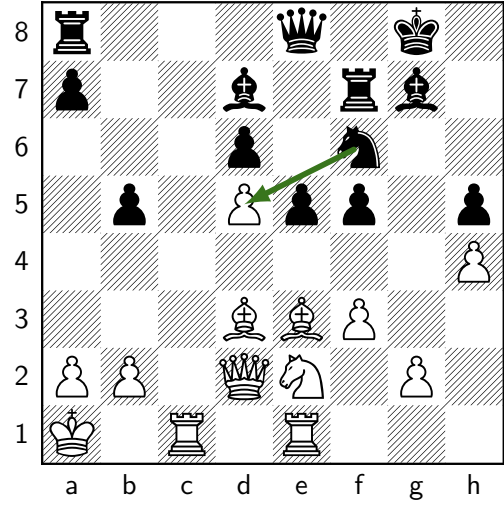


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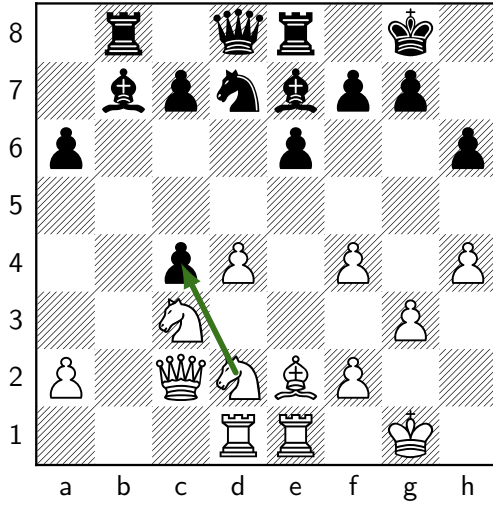
Figure 16: **Move 22, White: g3.** (a) Our position; Melanie plays g3 (g2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3398` (ply 32) [structural distance $d=88$, QAP obj=55]: the master played R_e1 ; under μ our $Pg2$ plays the role of that game’s $Rf1$ (a **cross-type** correspondence); cross-type roles: our $ph7$ as their $Bc1$, our $nb8$ as their $Qd1$; transferred move $g2g3$ (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2619` (ply 40) [structural distance $d=82$, QAP obj=47]: the master played $Bc3$; under μ our $Pg2$ plays the role of that game’s $Bb2$ (a **cross-type** correspondence); cross-type roles: our $bd7$ as their $Qd1$, our $nb8$ as their $Bc2$; transferred move $g2g3$ (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#480` (ply 43) [structural distance $d=78$, QAP obj=41]: the master played $Qe7$; under μ our $Pg2$ plays the role of that game’s $Qd1$ (a **cross-type** correspondence); cross-type roles: our $Pc5$ as their $Bb8$, our $Qd1$ as their $Re8$; transferred move $g2g3$ (**endorses this move**). Synthesis: 3 retrieved analogs transferred to g3 (summed QAP objective 143, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



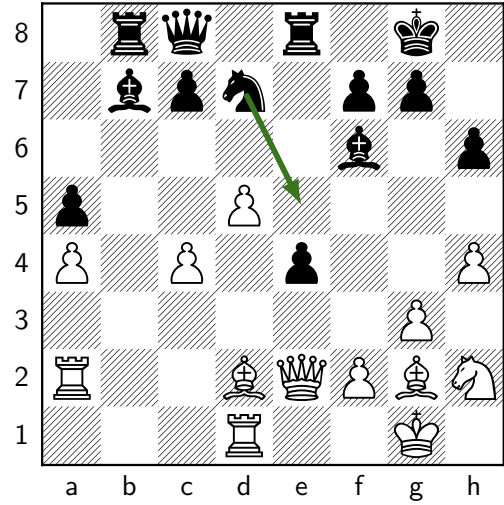
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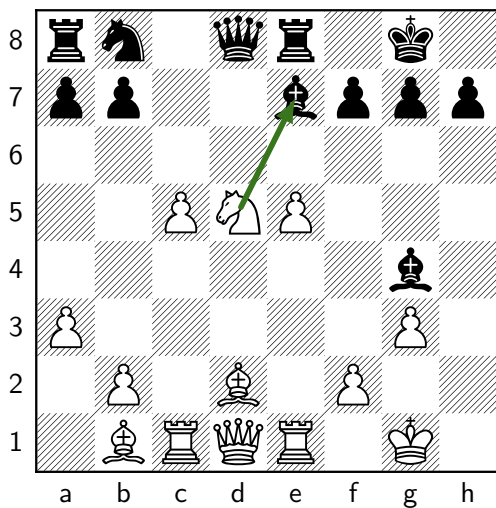


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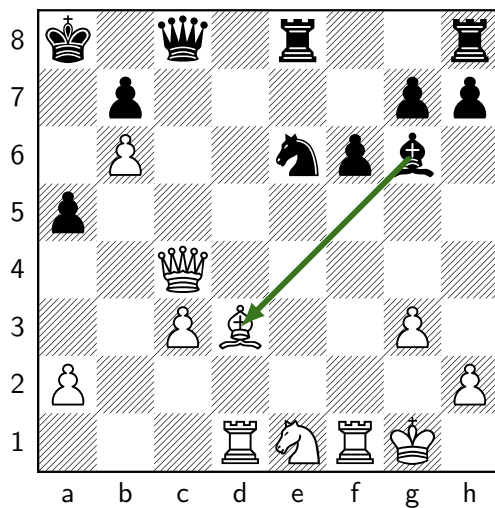


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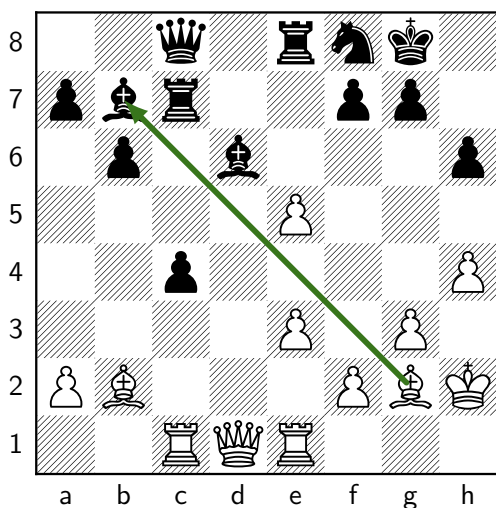
Figure 17: **Move 23, White: Nxd5.** (a) Our position; Melanie plays Nxd5 (c3d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#758` (ply 41) [structural distance $d=98$, QAP obj=73]: the master played Nxd5; under μ our *Nc3* plays the role of that game’s *Nf3*; cross-type roles: our *Pg3* as their *Ra8*, our *Re1* as their *Qe8*; transferred move c3d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3474` (ply 40) [structural distance $d=85$, QAP obj=51]: the master played Nxc4; under μ our *Nc3* plays the role of that game’s *Nd2*; cross-type roles: our *Qd1* as their *Rd1*, our *bd7* as their *Qc2*; transferred move c3d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1752` (ply 41) [structural distance $d=89$, QAP obj=85]: the master played Ne5; under μ our *Rc1* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *qd8* as their *Rb8*, our *Nc3* as their *Bf6*; transferred move c1c2. Synthesis: 2 retrieved analogs transferred to Nxd5 (summed QAP objective 124, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



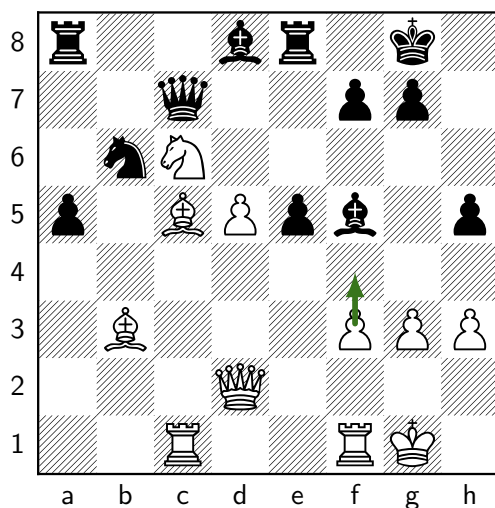
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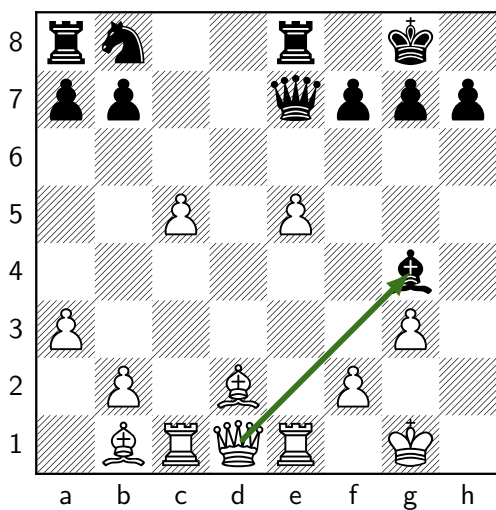


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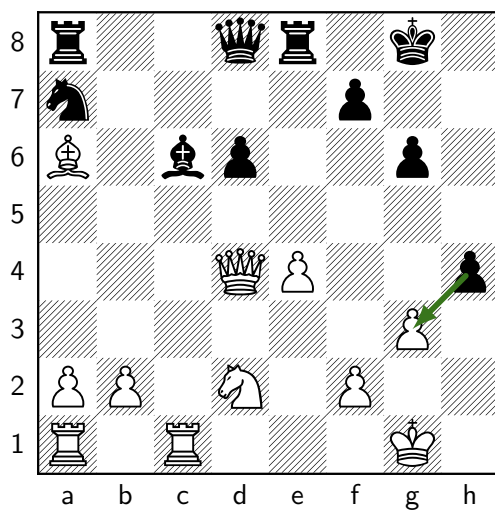


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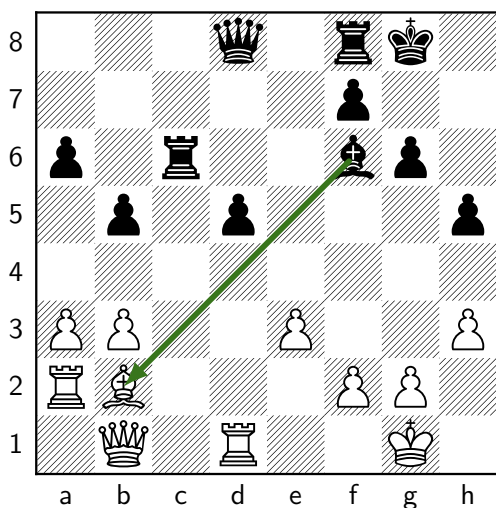
Figure 18: **Move 24, White: Nxe7+.** (a) Our position; Melanie plays Nxe7+ (d5e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#112` (ply 57) [structural distance $d=74$, QAP obj=36]: the master played Bxd3; under μ our Nd5 plays the role of that game’s Bg3 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Ne6, our bg4 as their qc4; transferred move d5e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3001` (ply 48) [structural distance $d=95$, QAP obj=30]: the master played Bxb7; under μ our Nd5 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our pb7 as their Re1, our be7 as their Pa2; transferred move d5e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2582` (ply 54) [structural distance $d=92$, QAP obj=50]: the master played f4; under μ our Pa3 plays the role of that game’s Pf3; cross-type roles: our Pe5 as their Rc1, our Bd2 as their Qd2; transferred move a3a4. Synthesis: 2 retrieved analogs transferred to Nxe7+ (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



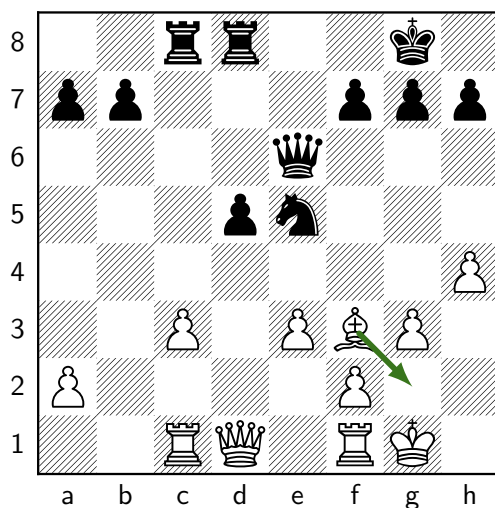
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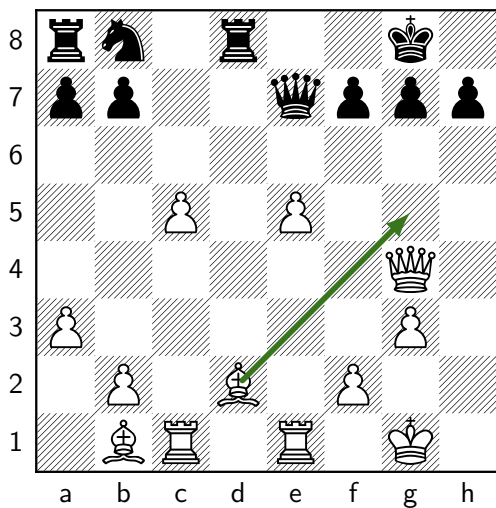


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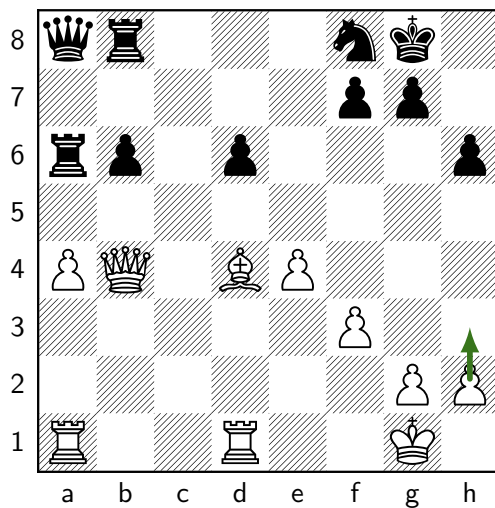


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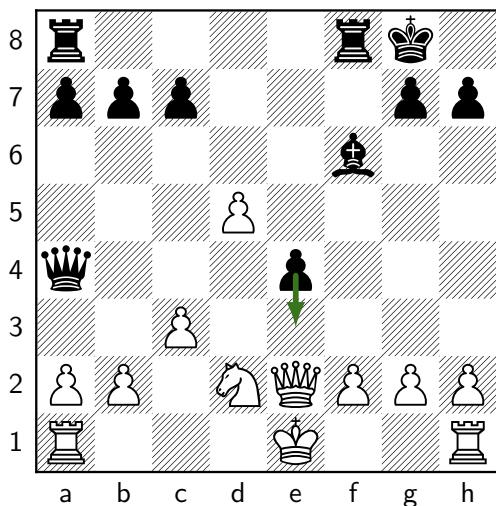
Figure 19: **Move 25, White: Qxg4.** (a) Our position; Melanie plays Qxg4 (d1g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1598` (ply 47) [structural distance $d=77$, QAP obj=48]: the master played **hxg3**; under μ our *Qd1* plays the role of that game’s *Ph5* (a **cross-type** correspondence); cross-type roles: our *Pf2* as their *Ra8*, our *Re1* as their *Qd8*; transferred move **d1g4 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#3827` (ply 45) [structural distance $d=70$, QAP obj=30]: the master played **Bxb2**; under μ our *Qd1* plays the role of that game’s *Bf3* (a **cross-type** correspondence); cross-type roles: our *Re1* as their *Qd8*, our *Bd2* as their *Rf8*; transferred move **d1g4 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#959` (ply 40) [structural distance $d=78$, QAP obj=26]: the master played **Bg2**; under μ our *Qd1* plays the role of that game’s *Bf3* (a **cross-type** correspondence); cross-type roles: our *ra8* as their *Pf2*, our *bg4* as their *ne5*; transferred move **d1g4 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to **Qxg4** (summed QAP objective 129, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



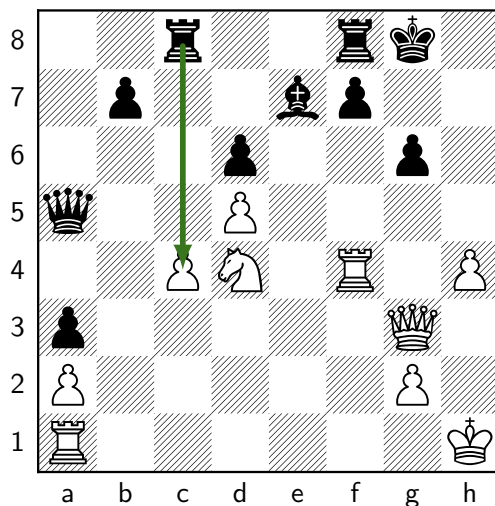
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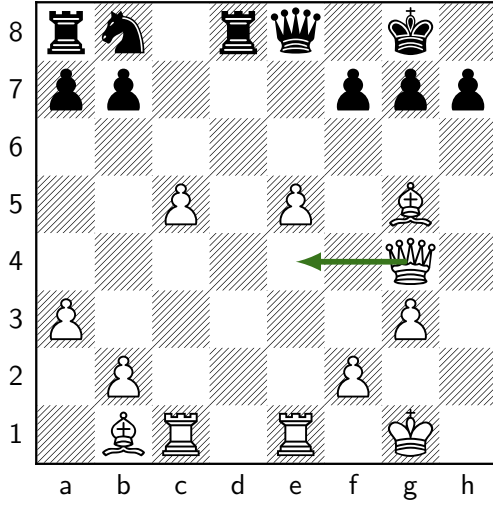


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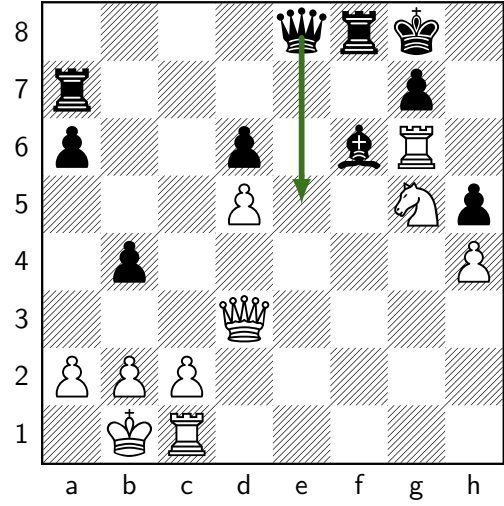


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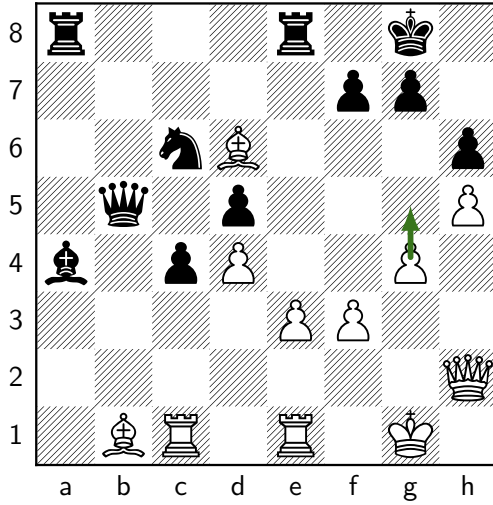
Figure 20: **Move 26, White: Bg5.** (a) Our position; Melanie plays Bg5 (d2g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3859` (ply 50) [structural distance $d=64$, QAP obj=140]: the master played h3; under μ our *Rc1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *pb7* as their *Ra1*, our *Bb1* as their *Pg2*; transferred move c1c2. (c) Retrieved from `ccr1_top_p0000#3830` (ply 35) [structural distance $d=64$, QAP obj=86]: the master played e3; under μ our *Bd2* plays the role of that game’s *Pe5* (a **cross-type** correspondence); cross-type roles: our *Bb1* as their *Pc7*, our *Qg4* as their *Bf6*; transferred move d2h6. (d) Retrieved from `ccr1_top_p0000#2935` (ply 59) [structural distance $d=52$, QAP obj=82]: the master played Rxc4; under μ our *Bb1* plays the role of that game’s *Rc1* (a **cross-type** correspondence); cross-type roles: our *pf7* as their *Rf8*, our *qe7* as their *Be7*; transferred move b1d3. Synthesis: no retrieved analog transferred to Bg5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



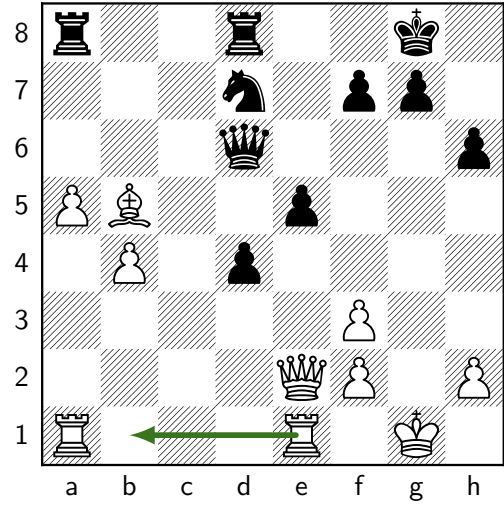
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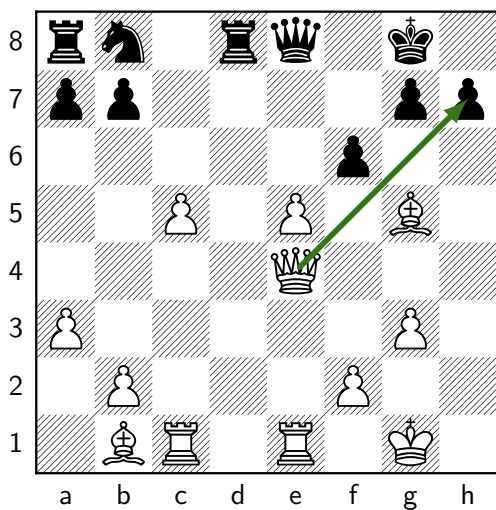


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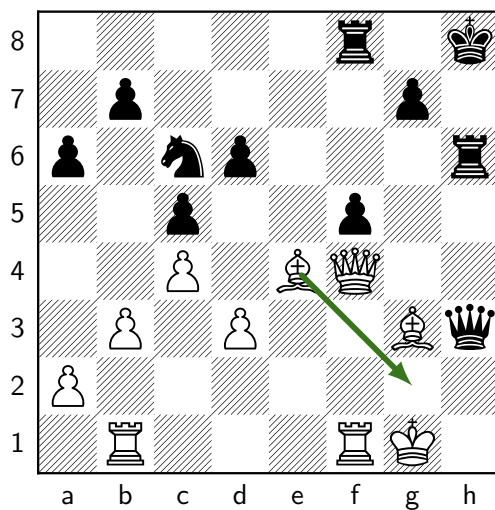


(d)

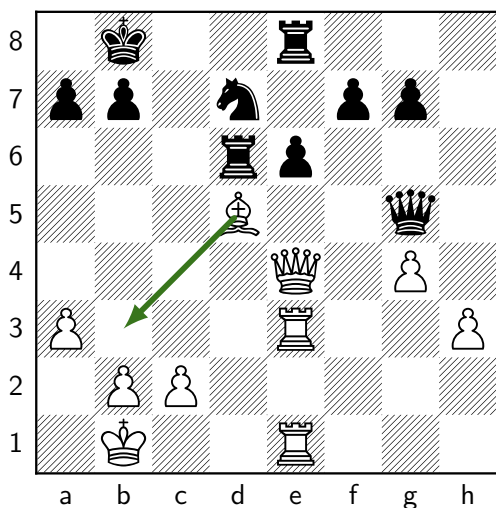
Figure 21: **Move 27, White: Qe4.** (a) Our position; Melanie plays Qe4 (g4e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2467` (ply 59) [structural distance $d=63$, QAP obj=21]: the master played Qe5; under μ our Qg4 plays the role of that game’s Qe1; cross-type roles: our qe8 as their rg6, our nb8 as their pd5; transferred move g4e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2867` (ply 62) [structural distance $d=76$, QAP obj=44]: the master played g5; under μ our Pb2 plays the role of that game’s Pg4; cross-type roles: our rd8 as their Bb1, our qe8 as their Rc1; transferred move b2b3. (d) Retrieved from `ccr1_top_p0001#2350` (ply 44) [structural distance $d=76$, QAP obj=42]: the master played Re1; under μ our Rc1 plays the role of that game’s Re1; cross-type roles: our Bb1 as their Ph2, our nb8 as their pe5; transferred move c1c2. Synthesis: 1 retrieved analog transferred to Qe4 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



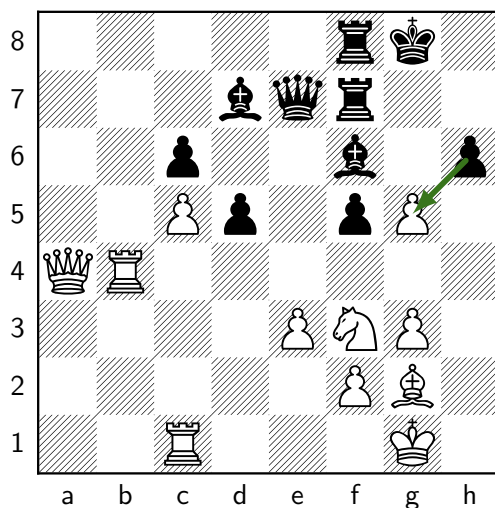
(a)



(b)

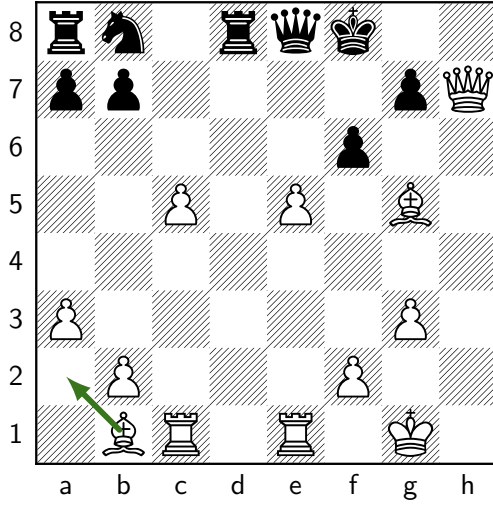


(c)

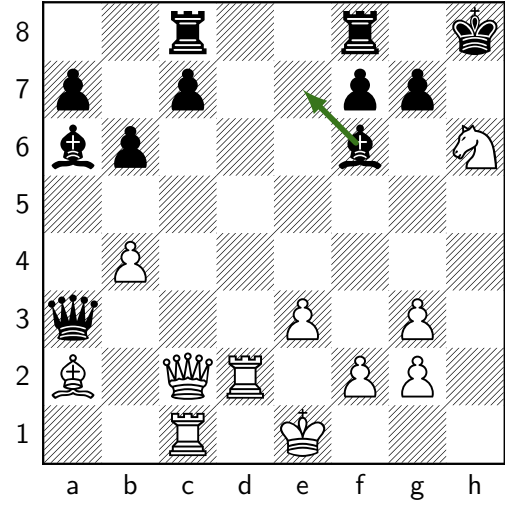


(d)

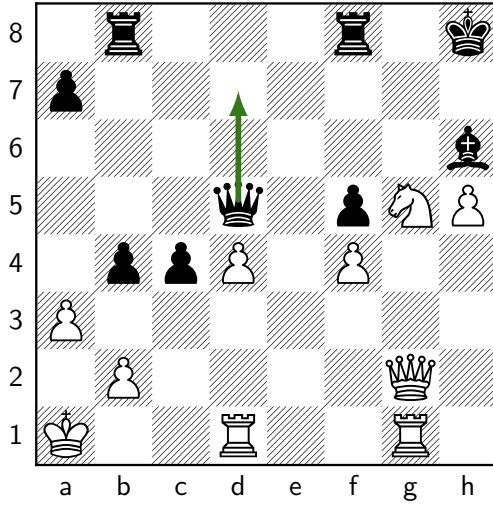
Figure 22: **Move 28, White: Qxh7+.** (a) Our position; Melanie plays Qxh7+ (e4h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3789` (ply 60) [structural distance $d=73$, QAP obj=71]: the master played Bg2; under μ our Bg5 plays the role of that game’s Be4; cross-type roles: our Bg5 as their Rf1, our pf6 as their qh3; transferred move g5d2. (c) Retrieved from `ccr1_top_p0000#3614` (ply 62) [structural distance $d=82$, QAP obj=70]: the master played Bb3; under μ our Bg5 plays the role of that game’s Bd5; cross-type roles: our Bb1 as their Pc2, our Bg5 as their Re3; transferred move g5h4. (d) Retrieved from `ccr1_top_p0002#292` (ply 61) [structural distance $d=79$, QAP obj=69]: the master played hxg5; under μ our Pe5 plays the role of that game’s Ph3; cross-type roles: our pg7 as their nf3, our nb8 as their pf2; transferred move e5f6. Synthesis: no retrieved analog transferred to Qxh7+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



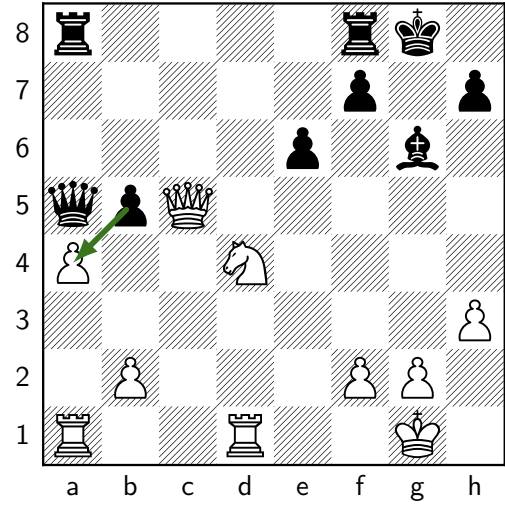
(a)



(b)

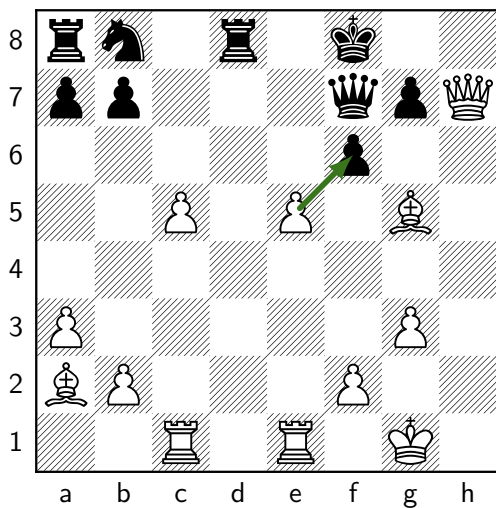


(c)

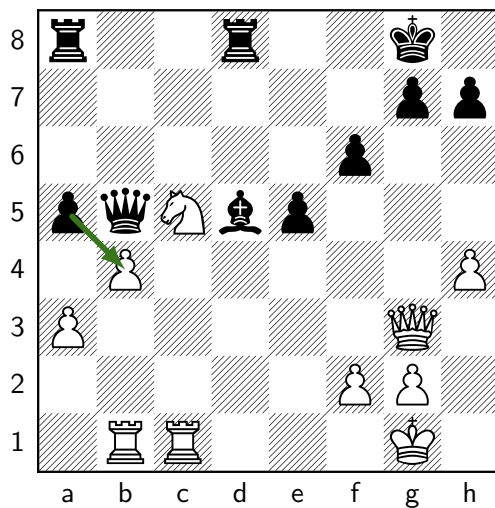


(d)

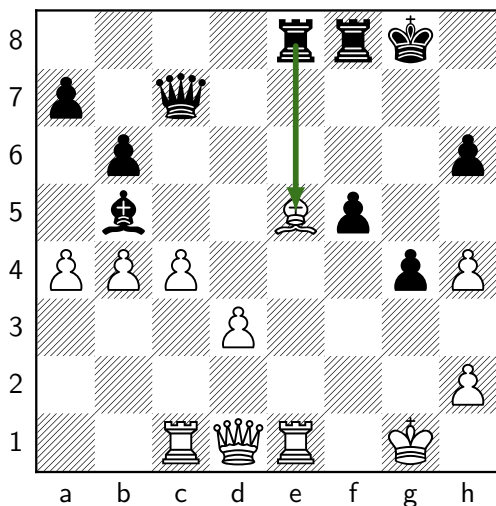
Figure 23: **Move 29, White: Ba2.** (a) Our position; Melanie plays Ba2 (b1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0002#1885` (ply 59) [structural distance $d=84$, QAP obj=53]: the master played Be7; under μ our Bb1 plays the role of that game’s Bf3; cross-type roles: our pg7 as their Rc8, our qe8 as their Pf7; transferred move b1a2 (**endorses this move**). (c) Retrieved from `ccr1.top_p0001#1486` (ply 61) [structural distance $d=78$, QAP obj=64]: the master played Qd7; under μ our Bg5 plays the role of that game’s Qd4 (a **cross-type** correspondence); cross-type roles: our pf6 as their Bh6, our Bg5 as their ng5; transferred move g5d2. (d) Retrieved from `ccr1.top_p0000#974` (ply 47) [structural distance $d=85$, QAP obj=60]: the master played bxa4; under μ our Pe5 plays the role of that game’s Pb4; cross-type roles: our Qh7 as their Ra8, our Pg3 as their Bg6; transferred move e5f6. Synthesis: 1 retrieved analog transferred to Ba2 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



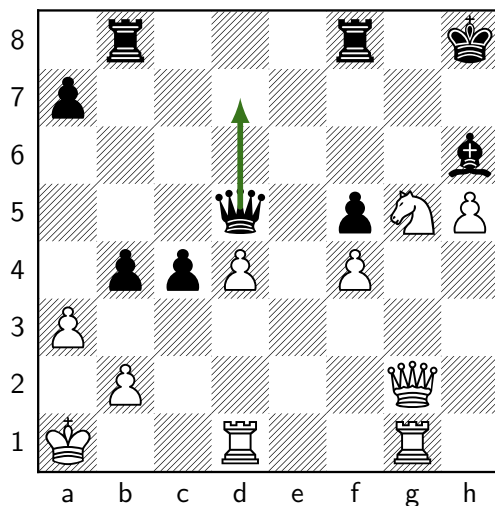
(a)



(b)

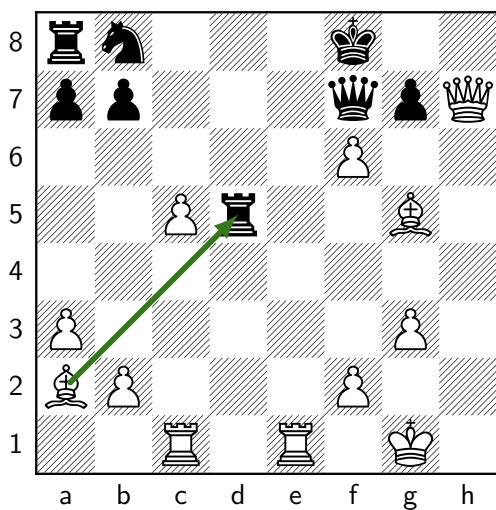


(c)

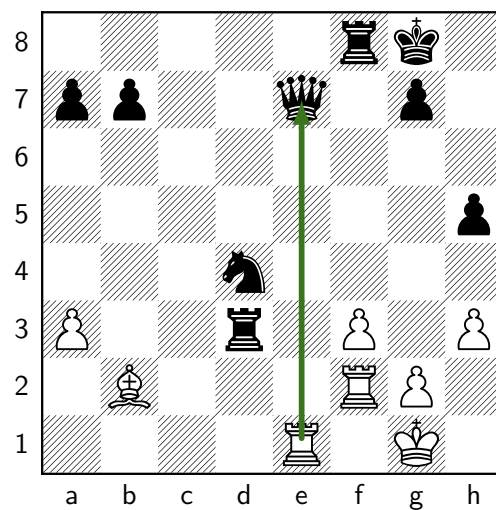


(d)

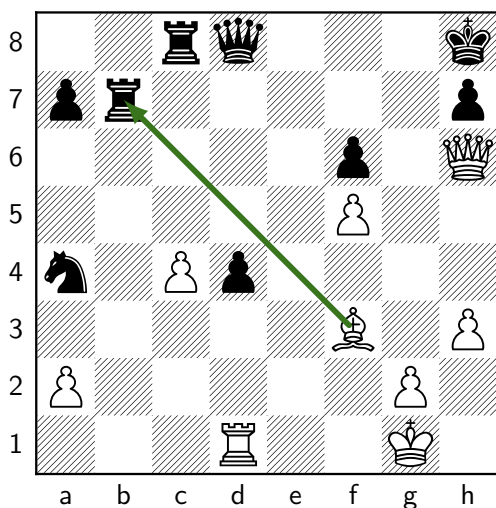
Figure 24: **Move 30, White: exf6.** (a) Our position; Melanie plays **exf6** (**e5f6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#206` (ply 57) [structural distance $d=73$, QAP obj=47]: the master played **axb4**; under μ our *Pe5* plays the role of that game’s *Pa4*; cross-type roles: our *Pg3* as their *Ra8*, our *Ba2* as their *Rd8*; transferred move **e5f6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1509` (ply 57) [structural distance $d=92$, QAP obj=67]: the master played **Rxe5**; under μ our *Bg5* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Rc1* as their *Pa7*, our *Bg5* as their *Qc7*; transferred move **g5h6**. (d) Retrieved from `ccr1_top_p0001#1486` (ply 61) [structural distance $d=86$, QAP obj=66]: the master played **Qd7**; under μ our *Bg5* plays the role of that game’s *Qd4* (a **cross-type** correspondence); cross-type roles: our *pg7* as their *Rb8*, our *Rc1* as their *Pf5*; transferred move **g5d2**. Synthesis: 1 retrieved analog transferred to **exf6** (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



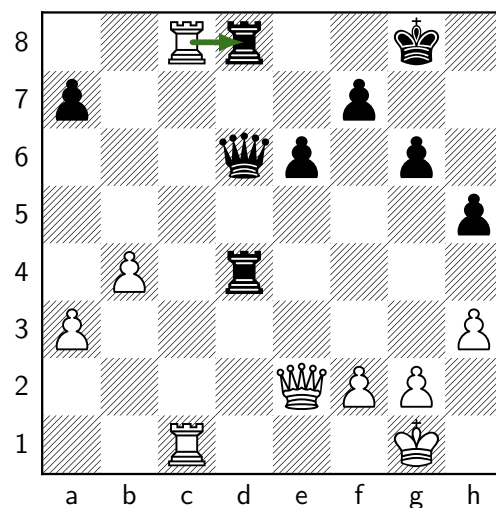
(a)



(b)

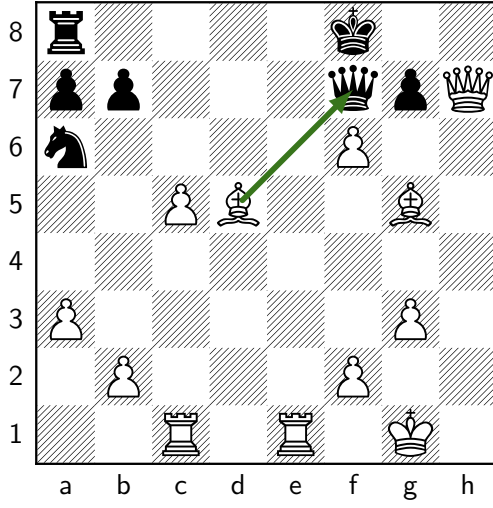


(c)

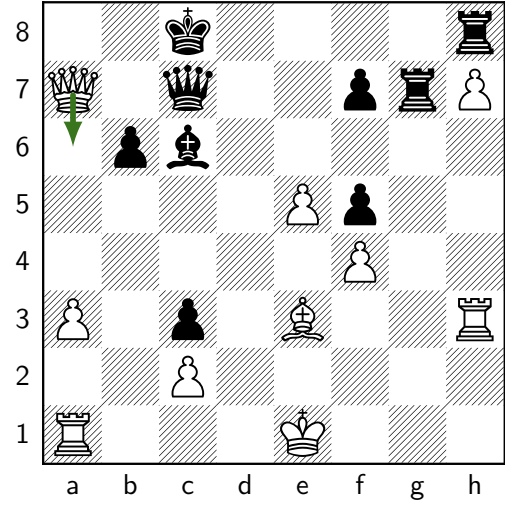


(d)

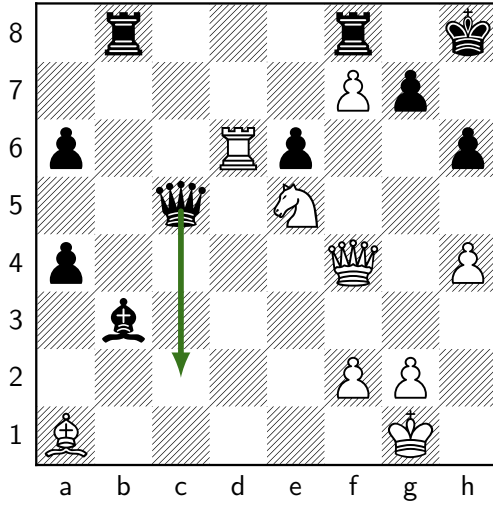
Figure 25: **Move 31, White: Bxd5.** (a) Our position; Melanie plays Bxd5 (a2d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2178` (ply 60) [structural distance $d=82$, QAP obj=46]: the master played Rxe7; under μ our Ba2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Qh7 as their Bb2, our Bg5 as their Rf2; transferred move a2d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2712` (ply 66) [structural distance $d=79$, QAP obj=43]: the master played Bxb7; under μ our Ba2 plays the role of that game’s Bf3; cross-type roles: our Rc1 as their Pg2, our Pa3 as their na4; transferred move a2d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#338` (ply 66) [structural distance $d=81$, QAP obj=21]: the master played Rxd8+; under μ our Ba2 plays the role of that game’s Rc8 (a **cross-type** correspondence); cross-type roles: our Bg5 as their Pa3, our Rc1 as their Pb4; transferred move a2d5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bxd5 (summed QAP objective 147, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



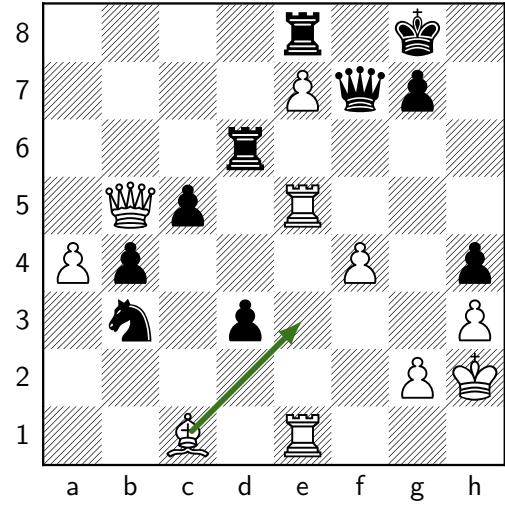
(a)



(b)

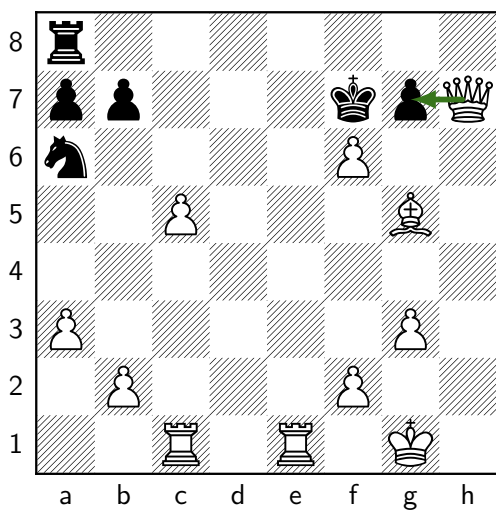


(c)

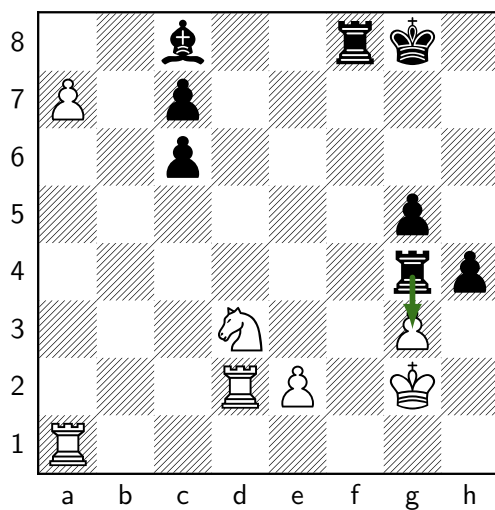


(d)

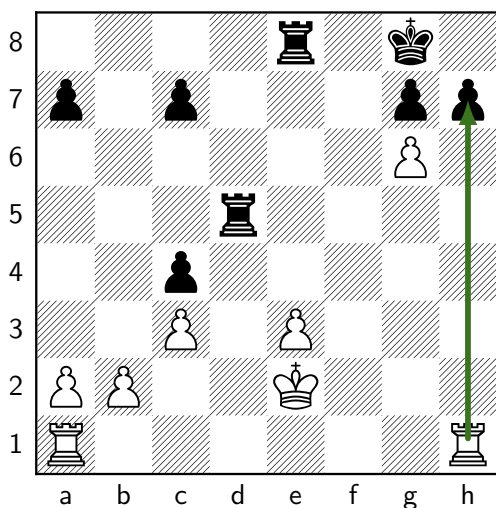
Figure 26: **Move 32, White: Bxf7.** (a) Our position; Melanie plays Bxf7 (d5f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#596` (ply 46) [structural distance $d=83$, QAP obj=70]: the master played Qa6+; under μ our Bd5 plays the role of that game’s Qa7 (a **cross-type** correspondence); cross-type roles: our ra8 as their Pa3, our na6 as their Be3; transferred move d5b7. (c) Retrieved from `ccr1_top_p0003#1576` (ply 65) [structural distance $d=84$, QAP obj=60]: the master played Qc2; under μ our Bd5 plays the role of that game’s Qc4 (a **cross-type** correspondence); cross-type roles: our Qh7 as their Rf8, our qf7 as their rd6; transferred move d5e4. (d) Retrieved from `ccr1_top_p0002#1508` (ply 72) [structural distance $d=83$, QAP obj=59]: the master played Be3; under μ our Bd5 plays the role of that game’s Bc1; cross-type roles: our Pa3 as their Re1, our qf7 as their nb3; transferred move d5e4. Synthesis: no retrieved analog transferred to Bxf7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



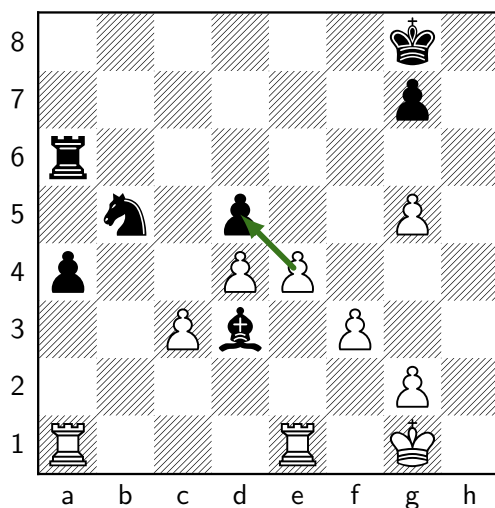
(a)



(b)



(c)



(d)

Figure 27: **Move 33, White: Qxg7#.** (a) Our position; Melanie plays Qxg7# (h7g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3104` (ply 71) [structural distance $d=62$, QAP obj=102]: the master played Rxg3+; under μ our Qh7 plays the role of that game’s Rg5 (a **cross-type** correspondence); cross-type roles: our Rc1 as their Bc8, our Bg5 as their Pg5; transferred move h7g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3830` (ply 54) [structural distance $d=65$, QAP obj=40]: the master played Rxxh7; under μ our Qh7 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Ra1, our na6 as their Pc3; transferred move h7g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#774` (ply 76) [structural distance $d=50$, QAP obj=82]: the master played exd5; under μ our Pf6 plays the role of that game’s Pe4; cross-type roles: our Qh7 as their Ra1, our kf7 as their bd3; transferred move f6g7. Synthesis: 2 retrieved analogs transferred to Qxg7# (summed QAP objective 142, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.